



Swami Keshvanand Institute of
Technology, Management & Gramothan,
Jaipur

I Mid Term Examination, Sept.-2022

Semester:	III	Branch:	CS/DS/AI/IT
Subject:	AEM	Subject Code:	3CS/AID/CAI/IT2-01
Time:	1.5 Hours	Maximum Marks:	20
Session (I/II/III): I			

PART A (short-answer type questions)

(All questions are compulsory)

(3*2=6)

Q.1 Determine the nature of the given matrix

$$A = \begin{bmatrix} -1 & -1 & -1 \\ -1 & -2 & -2 \\ -1 & -2 & -3 \end{bmatrix}$$

Q.2 A firm manufacturing two types of electric items A and B, can make a profit of Rs 20 per unit of A and Rs.30 per unit of B. Each unit of A requires 3 motors and 2 transformers and each unit of B requires 2 motors and 4 transformers. The total supply of these per month is restricted to 210 motors and 300 transformers. Type B is an export model requiring a voltage stabilizer which has a supply restricted to 65 units per month. Formulate it for maximize the profit.

Q.3 Write four application of engineering optimization.

PART B (Analytical/Problem solving questions)

(Attempt any 2 Questions)

(2*4=8)

Q.4 Determine the value of x for which the function $f(x) = x_1^2 + x_2^2 + x_3^2 - 4x_1 - 8x_2 - 12x_3 + 56$ assumes the optimum value.

Q.5 Write K.T. Conditions and then solve

Maximize $f = 10x_1 + 4x_2 - 2x_1^2 - x_2^2$ subject to $2x_1 + x_2 - 5 \leq 0$.

Q.6 Use Lagrange's multiplier method to solve

Maximize $z = 4x_1 - x_1^2 + 8x_2 - x_2^2$ subject to $x_1 + x_2 = 2, x_1, x_2 \geq 0$

PART C (Descriptive/Analytical/Problem solving/Design questions)

(Attempt any 1 Question)

(1*6=6)

Q.7 Solve by Simplex method *Maximum* $z = 5x_1 + 3x_2$
subject to $3x_1 + 5x_2 \leq 15$, $5x_1 + 2x_2 \leq 10$, $x_1, x_2 \geq 0$

Q.8 The estimated cost per unit time of five machines for five different jobs are given below. Solve the following assignment problem for minimum optimal cost

	1	2	3	4	5
A	12	8	7	15	4
B	7	9	17	14	10
C	9	6	12	6	7
D	7	6	14	6	10
E	9	6	12	10	7



Solution of Question Paper

I Mid-Term Examination, Sept. -2022

Branch/Semester: ...III... <u>CS, AI, DS, IT</u>	Subject: ... <u>AE, M</u>	Subject Code: <u>3CS2-01</u>
Duration: 1.5 hours	Date: <u>30/9/22</u> Session (I/II/III): ... <u>I</u> ...	Max Marks: ... <u>20</u>
Submitted By: <u>Dr. Shalini Shekhar</u>		

Ans. 1 If

$$A = \begin{bmatrix} -1 & -1 & -1 \\ -1 & -2 & -2 \\ -1 & -2 & -3 \end{bmatrix} \quad H_1 = -1, H_2 = 1, H_3 = -1$$

Negative definite

Ans. 2.

$$\text{Max. } Z = 20x + 30y$$

$$\begin{aligned} \text{Sub. to } 3x + 2y &\leq 210 \\ 2x + 4y &\leq 300 \\ y &\leq 65 \\ x, y &\geq 0 \end{aligned}$$

Ans 3. Engineering Application of Optimization

1. Inventory Control
2. Optimal production, planning, controlling & Scheduling.
3. Optimal design of electrical Networks.
4. Finding the optimal trajectories of space vehicles.

Ans 4.

$$\left. \begin{aligned} \frac{\partial f}{\partial x_1} &= 2x_1 - 4 = 0 \\ \frac{\partial f}{\partial x_2} &= 2x_2 - 8 = 0 \end{aligned} \right\} \frac{\partial f}{\partial x_3} = 0 \quad x_1 = 2, x_2 = 4, x_3 = 6$$

stationary point is (2, 4, 6)



Solution of Question Paper

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Branch/Semester: <u>CS/AI/DS/II III</u>	Subject: <u>ARM</u>	Subject Code: <u>3.CS/IT2-01</u>
Duration: 1.5 hours	Date: <u>30/9/22</u> Session (I/II/III): <u>I</u>	Max Marks: <u>20</u>
Submitted By:		

$$H(x) = \begin{bmatrix} \frac{\partial^2 f}{\partial x_1^2} & \frac{\partial^2 f}{\partial x_1 \partial x_2} & \frac{\partial^2 f}{\partial x_1 \partial x_3} \\ \frac{\partial^2 f}{\partial x_2 \partial x_1} & \frac{\partial^2 f}{\partial x_2^2} & \frac{\partial^2 f}{\partial x_2 \partial x_3} \\ \frac{\partial^2 f}{\partial x_3 \partial x_1} & \frac{\partial^2 f}{\partial x_3 \partial x_2} & \frac{\partial^2 f}{\partial x_3^2} \end{bmatrix} = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

$$H_1 = 2, H_2 = 4, H_3 = 8$$

Since all are positive so function is minimum at point (2, 4, 6).

Ans. 5 The k.T. Conditions are

$$\frac{\partial f}{\partial x_j} - \lambda \frac{\partial g}{\partial x_j} = 0, \lambda g(x) = 0, g(x) \leq 0, \lambda \geq 0$$

Using these conditions we get

$$10 - 4x_1 - 2\lambda = 0 \quad \text{--- (1)}$$

$$4 - 2x_2 - \lambda = 0 \quad \text{--- (2)}$$

$$\lambda(2x_1 + x_2 - 5) = 0 \quad \text{--- (3)}$$

$$2x_1 + x_2 - 5 \leq 0 \quad \text{--- (4)}$$

$$\lambda \geq 0 \quad \text{--- (5)}$$

from (3) either $\lambda = 0$ or $2x_1 + x_2 - 5 = 0$

Let $\lambda = 0$ then $x_1 = 2.5, x_2 = 2$ which is not satisfying eqn. (4)

So we assume $2x_1 + x_2 - 5 = 0, \lambda \neq 0$



Solution of Question Paper
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Branch/Semester: <u>CS/AI/DS/IT III</u>	Subject: <u>AGM</u>	Subject Code: <u>3CS/IT2-01</u>
Duration: 1.5 hours	Date: <u>30/9/22</u> Session (I/II/III): <u>I</u>	Max Marks: <u>20</u>
Submitted By:		

we have $\lambda = \frac{4}{3}$, $x_2 = \frac{4}{3}$, $x_1 = \frac{11}{6}$, which satisfy all the equations, Hence $x_1 = \frac{11}{6}$, $x_2 = \frac{4}{3}$, $z_{\max} = \frac{91}{6}$.

Ans. 6 Let the lagrangian be

$$L(x_1, x_2, \lambda) = 4x_1 - x_1^2 + 8x_2 - x_2^2 - \lambda(x_1 + x_2 - 2)$$

$$\frac{\partial L}{\partial x_1} = 4 - 2x_1 - \lambda = 0 \quad \text{--- (1)}$$

$$\frac{\partial L}{\partial x_2} = 8 - 2x_2 - \lambda = 0 \quad \text{--- (2)}$$

$$\frac{\partial L}{\partial \lambda} = x_1 + x_2 - 2 = 0 \quad \text{--- (3)}$$

$\Rightarrow$ on solving we get

$$x_1 = 0, x_2 = 2, \lambda = 4$$

Now
$$H = \begin{bmatrix} \frac{\partial^2 L}{\partial x_1^2} - \lambda & \frac{\partial^2 L}{\partial x_1 \partial x_2} \\ \frac{\partial^2 L}{\partial x_2 \partial x_1} & \frac{\partial^2 L}{\partial x_2^2} - \lambda \end{bmatrix} = \begin{bmatrix} -2 - \lambda & 0 \\ 0 & -2 - \lambda \end{bmatrix} = 0$$

$$\Rightarrow \lambda = -2$$

Since eigen value is negative so the function is maximum at $x_1 = 0, x_2 = 2, z_{\max} = 12$

Ans. 7

The standard form of LPP is

$$\text{Max } z = 5x_1 + 3x_2 + 0x_3 + 0x_4$$

$$\text{sub. to } 3x_1 + 5x_2 + x_3 + 0x_4 = 15$$

$$5x_1 + 2x_2 + 0x_3 + x_4 = 10$$

$$x_1, x_2, x_3, x_4 \geq 0$$



Solution of Question Paper

I Mid-Term Examination, Sept. -2022

Branch/Semester: <u>CS/IT/DS/AI II</u>	Subject: <u>ARM</u>	Subject Code: <u>36S/IT2-01</u>
Duration: 1.5 hours	Date: <u>3.9.22</u> Session (I/II/III): <u>I</u>	Max Marks: <u>20</u>
Submitted By:		

C_B	B	x_j	b_j	y_1	y_2	y_3	y_4	Min Ratio
0	x_3	15	3	5	1	0		$15/3 = 5$
0	x_4	10	5	2	0	1		$10/5 = 2$
$Z_j - C_j$			-5	-3	0	0		
0	x_3	9	0	$19/5$	1	$-3/5$		$45/19$
5	x_1	2	1	$2/5$	0	$1/5$		$10/2$
$Z_j - C_j$			0	-1	0	1		
3	x_2	$45/19$	0	1	$5/19$	$-3/19$		
5	x_1	$20/19$	1	0	$-2/19$	$5/19$		
$Z_j - C_j$			0	0	$5/19$	$16/19$		

$\text{All } Z_j - C_j \geq 0$
 The optimal solution is
 $x_1 = \frac{20}{19}, x_2 = \frac{45}{19}$
 $\text{Max. } Z = \frac{235}{19}$

Ans. 8.

Step I

	1	2	3	4	5
I	8	4	3	11	0
II	0	2	10	7	3
III	3	0	6	0	1
IV	1	0	8	0	4
V	3	0	6	4	1

	1	2	3	4	5
I	8	4	0	11	0
II	0	2	7	7	3
III	3	0	3	0	1
IV	1	0	5	0	4
V	3	0	3	4	1

	1	2	3	4	5
I	8	5	0	12	0
II	0	3	7	8	3
III	2	0	2	0	0
IV	0	0	4	0	3
V	2	0	2	4	0

I $\rightarrow$ 3

II $\rightarrow$ 1

III $\rightarrow$ 4

IV $\rightarrow$ 2

V $\rightarrow$ 5

Min Cost

$$= 7 + 7 + 6 + 6 + 7 = 33.$$



**Swami Keshvanand Institute of Technology,
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I Mid Term Examination, Sept.-2022

Semester:	III	Branch:	CS/AI/DS
Subject:	Technical Communication	Subject Code:	3CS/AI/DS 1-02
Time:	1.5 Hours	Maximum Marks:	20
Session:	II		

PART-A (All questions are compulsory)

3*2=6

- Q1. Define Technical Communication.
Q2. What are the ABCS of Technical Communication?
Q3. Create the format of an email?

PART-B (Attempt any two questions)

2*4=8

- Q4. Describe the strategies for reading texts.
Q5. Write a short note on aspects/ competencies of technical communication.
Q6. Explain any two methods of note-making and draw diagrams to support your answer. Which is your favourite method?

PART-C (Attempt any one question)

1*6=6

- Q7. Define Project proposal and describe the types of proposals.

Q-8. Do as directed:

- You ----- pay your taxes in time. (Shall/should/might/ must).
- Although I was hungry, I didn't stop studying. (Correct the error, if any)
- Who stole my cheese? (Change into passive voice)
- I won't go out, as it was raining now and I do not have an umbrella.
(Correct the error, if any)
- It has been raining for 5 o'clock, so we stood under a shade. (Correct the error, if any)
- He ----- swim when he was four years old (can/ could/ will)



Solution of Question Paper
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Branch/Semester: CS/AI/DS --III	Subject: Technical Communication	Subject Code: 3CS/AI/DS 1-2
Duration: 1.5 hours	Date:..... Session (I/II/III): II	Max Marks:20
Submitted By: Krishna Sharma		

PART -A

Q-1 Definition- Technical communication can be considered as a medium to communicate scientific knowledge, engineering concepts and other technical information.

Society of Technical Communication defines the field as “any field of communication that focuses on the specialized topics, communicates specifically by using technology or providing instructions on how to do something.”

The European Association for Technical Communication briefly explores technical communication as “the process of defining, creating and exploring information for the safe, efficient use of products (technical systems, software services.)

Q-2. The acronym ABCS in technical communication refers to:

Accuracy: Accurate information of facts and figures while reporting, presentation etc. It refers to providing specific information rather than generalisations.

Brevity: Try to keep the document as short as possible

Clarity: Use of familiar vocabulary and constructions and being consistent with your terminology, abbreviations and presentation of figures, tables, illustrations, etc.

Simplicity: This is about expressing our thoughts with simplicity, not simplifying our work. It refers to the use of simple familiar words and phrase according to the standards of target audience

Q.3. Email Writing Format

To: (email id of recipient)

Cc:

Bcc:

Subject: -----

Salutation: -----

Body: -----

Thanks & regards

(Your name/ organisation)

PART -B

Q. 4. Reading is a learning skill. It helps you improve all parts of the English language – vocabulary, spelling, grammar, and writing.

It helps to develop language intuition in the corrected form. Then the brain imitates them, producing similar sentences to express the desired meaning.

Using skimming or scanning technique to read quickly is highly effective.

While reading underlining of key words is a must.

There are various techniques and strategies for reading any text. The purpose of reading a text must be identified first whether it is reading for exams or for entertainment. Proper prediction techniques can be used to infer what the text is about.



Solution of Question Paper

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Submitted By: Krishna Sharma		

There four important techniques are

1. **Skimming**—it is reading a text quickly to get a general idea or gist of the text (theme, topic or meaning. It is useful for pre-reading tasks or texts that do not require detailed attention. There is no need to read every word, the focus is on improving the speed of reading. This technique has lower comprehension of details.
2. **Scanning**-- provides some key information about the text. It is reading the text quickly to find out specific information e.g., finding the name or figures in a list/ chart. It is useful for answering questions which ask for true or false. The chief objective of scanning is to search out relevant words, collocations etc. the focus is more on lexis and to ignore irrelevant information.
3. **Intensive Reading**— (Text books)—refers to reading shorter texts for specific purpose wherein the learners are supposed to read the text carefully and deeply in order to gain maximum understanding, for instance reading for exam or delivering a speech etc. The reader concentrates on vocabulary and grammar for better comprehension of the text.
4. **Extensive Reading** (novels, magazines, news papers etc.) – refers to reading longer text for pleasure to develop reading skills.

There are also some more reading techniques like

- **KWL Technique**— Students write this list in the “K” column of their K-W-L forms.

The acronym stands for

1. What I Know
2. What I Want to Know
3. What I Learnt

- **SQ3R Technique**

Survey- the text quickly to discover its purpose and identify its organization, layout, chapters, images etc.

Question—Ask appropriate questions for each part of the text for better understanding.

Read—carefully, thoroughly- making notes

Recall—the content of each part of the text and amend the noted if required.

Revise—check accuracy of reading recall and review the notes.

- **ERRQ Method**—

Estimate—and predict what the text will be like. Use rapid reading (skimming) for this

Read—the text carefully

Respond- to the text, analyze what you know, want to know and what you learnt.



Solution of Question Paper

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Submitted By: Krishna Sharma		

Question-- ask questions for better comprehension

Q-5 Aspects of Technical Communication

Effective communication is a dynamic interchange that requires a systematic understanding of technical and scientific subjects.

According to M. A. Rizvi, the four major requirements of effective technical communication are:

1. Subject/ Content competence
2. Linguistic competence
3. Skills competence
4. Organisational competence

Subject Competence

Ideation or encoding in the technical communication process depends on the sender's subject or content competence. The sender should be equipped with sufficient knowledge of the particular subject matter and must possess highly sophisticated technical or professional skills. Ignorance of the subject knowledge and skills might render the communication ineffective.

Linguistic Competence

Linguistic competence refers to possession of appropriate language skills to present the technical facts/information in a clear and precise manner. Technical communication involves technical presentation of data in reports, proposals, research papers, bulletins, manuals and handbooks etc.

These skills include the ability to:

- Analyse facts or information for a clear meaningful presentation.
- Use appropriate rhetorical devices to present the scientific data.
- Use visual aids like charts, graphs, pictures and diagrams systematically

Skills Competence

Skills competence refers to possession of computer related presentation and technical skills. Being techno-savvy and having knowledge about the latest technical and communication skills keeps one updated in that area. Ignorance or lack of these skills might lead to ineffective communication.

Some skills required for Technical Communication include

- Professional Skills



Solution of Question Paper

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Submitted By: Krishna Sharma		

- Advanced writing and editing
- Research and analysis
- Technical knowledge comprehension
- Document design
- Digital and technical media
- Visual communication
- Teamwork and collaboration
- Organization and project management

Organizational Competence

Since technical communication is a systematic and structured presentation of facts and information, a thematic and logical organisation becomes indispensable. It includes the ability to

- Present the thought in a sequence
- Organise sentences/ paragraphs according to the needs of the subject and audience.
- Use logical, thematic and chronological order to provide coherence.

Q.6 Note making is not just about writing down everything you hear or read. It is a process of reviewing, connecting and synthesizing ideas from your lectures or reading. **Note Making** is a way of recording important details from a source. This source can be any book, article, meeting or any oral discussion. In note making, the writer records the essence of the information. It helps us to understand and clarify thinking. Note making saves a lot of time by going through the notes made. One can get a glimpse of a lot of information from a short note.

There are various methods of note-making like the Outline Format/ Method, the Cornell Format/ method, the Charting Format, the Mind mapping/ Bubble mapping Format, the Sentence Format, the Next Generation Note-taking method (e.g. Fishtree E-notes). My favourite methods is the outline method.

A. The Outline Format: this is a very popular method used by students. In this method notes are written in the form of outlines by using bullet points/ numbers. This method saves time for further reviewing and editing thus making the notes handy. The main topics are written on the extreme left of the page and related subtopics are added as bullet points.

Advantages:

it highlights the key points in a logical manner
it reduces the reviewing and editing time.

Disadvantages:

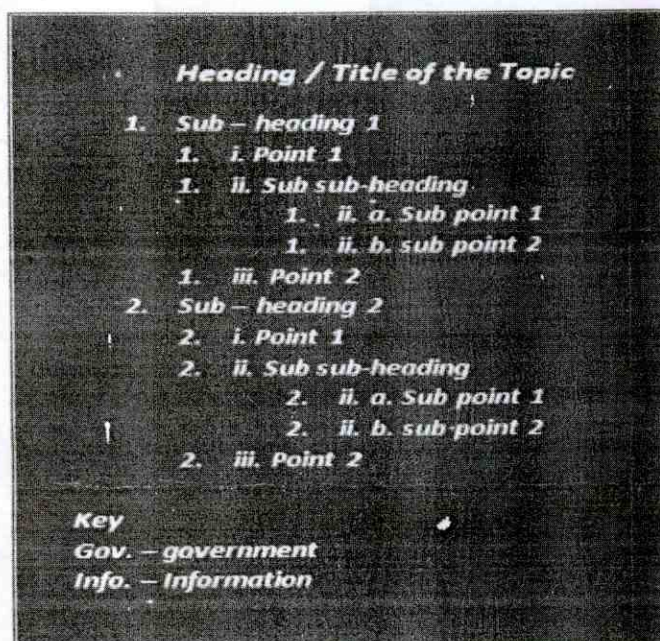


Solution of Question Paper

I Mid-Term Examination, Sept. -2022

Branch/Semester: CS/AI/DS --III	Subject: Technical Communication	Subject Code: 3CS/AI/DS 1-2
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Not suitable for subjects like chemistry, mathematics that comprise of formulas and charts.



B. The Cornell Format:

In this system the page is divided into two columns. The note taking column (usually on the right) is twice the size of the Key words/ cues column (on the left). A vertical line/ margin is drawn two and a half inches from the left side of the paper. This is known as recall column. All the lecture notes are taken on the right of this margin. The keywords/ cues, comments, questions are written in the left (recall) column. At the bottom of the page about 5-7 lines space is left to summarize main ideas to help in reviewing and speeding up the studying process.

Advantages:

It summarizes information in a systematic manner
It helps in quick and efficient learning in a short time.

Disadvantages:

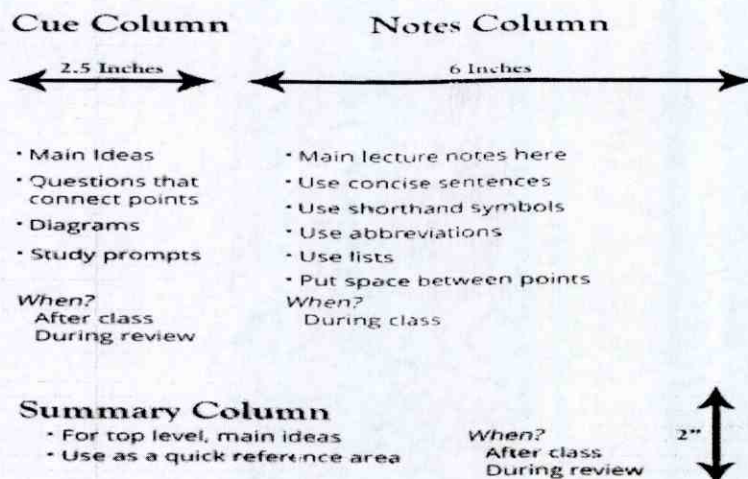
The format has to be prepared prior to taking notes
It requires some time for reviewing and summarizing the key concepts.



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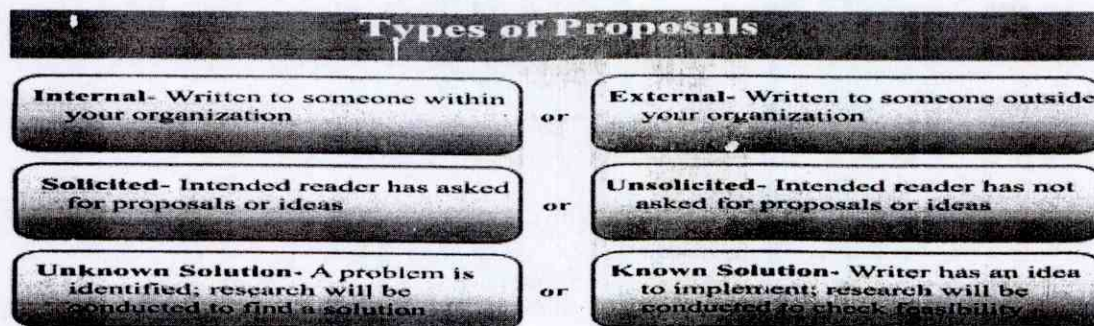
Cornell Notetaking Method



Q. 7 A proposal is a method of persuading people to agree with the proposer's views or accept his suggestions.

Proposal is a systematic, factual and persuasive description of a course of action or a set of recommendations/suggestions. It is written for a specific audience to meet specific needs. It explains and justifies what it proposes.

The Project Proposal is the initial document used to define an internal or external project. The proposal includes sections such as title, start and end dates, objectives and goals, requirements, and a description of the proposed solution.





Solution of Question Paper

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Submitted By: Krishna Sharma		

Types of proposals are:

1. On the basis of content and format --Non-formal and Formal Proposals
2. On the basis of Nature of audience --- Internal and External Proposals
3. On the basis of source of origin --- Solicited and Unsolicited Proposals

Non-formal and Formal Proposals (On the basis of content and format):

Non-formal proposal is a brief description of suggestions or recommendations that are introductory in nature. They are usually short and involve the use printed forms, letter formats or memo formats.

Formal proposals on the other hand are comparatively longer. They are usually written to initiate big projects and require elaborate descriptions and discussions. It may consist of various sections and sub-sections and can vary from a few pages to hundreds of pages. Like a formal report, a formal proposal involves the use of manuscript format

Internal and External Proposals (on the basis of Nature of audience):

According to the nature of the audience, the proposals may be internal or external

A proposal to someone within your **organization (a business, a government agency, etc.)** is an internal proposal. **An internal proposal is addressed to the readers within an organisation.**

It may offer to study problem, situation or issue within an organization or company accompanied with recommendations and suggestions to solve them. E.g. proposing a plan to increase sales. The internal proposals are less formal and elaborate.

An external proposal is one written from one separate, independent organization or individual to another such entity. The typical example is the independent consultant proposing to do a project for another firm. External proposals are more formal and elaborate.

Solicited and Unsolicited Proposal (on the basis of source of origin):

A solicited proposal is one in which the recipient has requested the proposal. Typically, a company will



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send out **requests for proposals (RFPs)** through the mail or publish them in some news source. But proposals can be solicited on a very local level: for example, you could be explaining to your boss what a great thing it would be to install a new technology in the office; your boss might get interested and ask you to write up a proposal that offered to do a formal study of the idea. **Solicited proposals are written as a response to a specific request. Many companies, government agencies and consultancy organizations solicit proposal for their projects.**

Unsolicited proposals are written without any request for a proposal. In unsolicited proposals the recipient has not requested proposals. With unsolicited proposals, you sometimes must convince the recipient that a problem or need exists before you can begin the main part of the proposal.

Q-8. Do as directed:

- You **must** ---- pay your taxes in time.
- Although I was hungry, (but) I didn't stop studying.
- By whom** was my cheese stolen?
- I won't go out, as it is **raining** now and I **do not have** an umbrella.
- It has been raining **since** 5 o'clock, so we stood **under** a shade.
- He **could** swim when he was four years old.



**Swami Keshvanand Institute of Technology,
Management & Gramothan, Jaipur
I Mid Term Examination, Sept.-2022**

Semester:	III	Branch:	CSE, AI, DS
Subject:	DE	Subject Code:	3CAI3-04, 3CS3-04, 3DS3-04
Time:	1.5 Hours	Maximum Marks:	20
Session (I/II/III): I			

PART A (short-answer type questions)

(All questions are compulsory)

(3*2=6)

Q.1 Solve the following:

- (a) Octal equivalent of $1BC_{16}$
- (b) Binary equivalent of 174_{10}
- (c) BCD and decimal equivalent of $(1001001)_{XS-3}$

PART B (Analytical/Problem solving questions)

(Attempt any 2 Questions)

(2*4=8)

Q.2 Show that in consensus law $(A+B)(A'+C)(B+C) = (A+B)(A'+C)$ and $AB + A'C + BC = AB + A'C$

Q.3 Explain the following terms: prime implicants, essential prime implicants, redundant prime implicants and selective prime implicants.

Q.4 Describe 2's complement subtraction procedure and show the subtraction of $(10110)_2 - (11010)_2$.

PART C (Descriptive/Analytical/Problem solving/Design questions)

(Attempt any 1 Question)

(1*6=6)

Q.5 Illustrate the following function in

(a) SOP minimal form using K-Maps:

$$F(A, B, C, D) = \sum_m(1, 2, 6, 7, 8, 13, 14, 15) + d(3, 5, 12)$$

(b) POS minimal form using K-Maps:

$$F(A, B, C, D) = \prod_M(1, 5, 6, 12, 13, 15) + d(2, 4)$$

Q.6 Determine the minimal sum of products for the Boolean expressions

$$F(A, B, C, D) = \sum_m(2, 6, 8, 9, 10, 11, 14, 15) \text{ using the Quine-McCluskey method.}$$



Solution of Question Paper
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Branch/Semester: III	Subject: DE	Subject Code: 3CAI3-04, 3CS3-04, 3DS3-04
Duration: 1.5 hours	Date: 28.09.2022 Session (I/II/III): I	Max Marks: 20
Submitted By: Shanu Tripathi, Dr. Vinay Kanungo, Dr. Niketa Sharma, Vinod Kataria		

Q.1 Solve the following:

(a) Octal equivalent of $1BC_{16}$

Ans1(a) Given, $1BC_{16}$ is a hexadecimal number.

$1 \rightarrow 0001, B \rightarrow 1011, C \rightarrow 1100$

Now group them from right to left, each having 3 digits.

000, 110, 111, 100

$000 \rightarrow 0, 110 \rightarrow 6, 111 \rightarrow 7, 100 \rightarrow 4$

Hence, $1BC_{16} = 674_8$

(b) Binary equivalent of 174_{10}

Division by 2	Quotient	Remainder	Bit #
$174/2$	87	0	0
$87/2$	43	1	1
$43/2$	21	1	2
$21/2$	10	1	3
$10/2$	5	0	4
$5/2$	2	1	5
$2/2$	1	0	6
$1/2$	0	1	7

So $174_{10} = 10101110_2$

(c) BCD and decimal equivalent of $(1001001)_{XS-3}$

So, grouping 4-bit for each group, i.e., 0100 1001 and subtract 0011 0011 from given number. Therefore,

$= 0100\ 1001 - 0011\ 0011 = 0001\ 0110$

So, binary coded decimal number is 0001 0110 and decimal number will be 16



Solution of Question Paper

I Mid-Term Examination, Sept. -2022

Branch/Semester: III	Subject: DE	Subject Code: 3CAI3-04, 3CS3-04, 3DS3-04
Duration: 1.5 hours	Date: 28.09.2022 Session (I/II/III): I	Max Marks: 20
Submitted By: Shanu Tripathi, Dr. Vinay Kanungo, Dr. Niketa Sharma, Vinod Kataria		

Q.2 Show that in consensus law $(A+B)(A'+C)(B+C) = (A+B)(A'+C)$ and $AB + A'C + BC = AB + A'C$
Ans: $(A+B)(A'+C)(B+C) = (A+B)(A'+C)$

$$\begin{aligned}
 \text{LHS} &= (A+B)(B'+C)(A+C) \\
 &= (A+B)(B'+C)(A+C+0) \\
 &= (A+B)(B'+C)(A+B.B'+C) \\
 &= (A+B)(B'+C)(A+B+C)(A+B'+C) \\
 &= [(A+B)(A+B+C)][(B'+C)(A+B'+C)] \\
 &= (A+B)(B'+C) = \text{RHS}
 \end{aligned}$$

$$AB + A'C + BC = AB + A'C$$

$$Y = AB + A'C + BC$$

$$Y = AB + A'C + BC.1$$

$$Y = AB + A'C + BC.(A+A')$$

$$Y = AB + A'C + ABC + A'BC$$

$$Y = AB(1+C) + A'C(1+B)$$

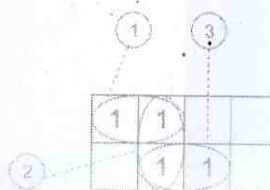
$$Y = AB + A'C$$

Q.3 Explain the following terms: prime implicants, essential prime implicants, redundant prime implicants and selective prime implicants.

Ans 3.

Prime Implicants – A group of square or rectangle made up of bunch of adjacent minterms which is allowed by definition of K-Map are called **prime implicants(PI)** i.e. all possible groups formed in K-Map.

Example:



No. of Prime Implicants = 3

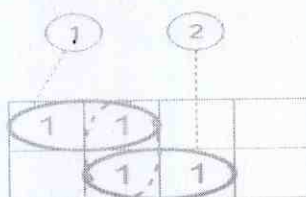


Solution of Question Paper
I Mid-Term Examination, Sept. -2022

Branch/Semester: III	Subject: DE	Subject Code: 3CAI3-04, 3CS3-04, 3DS3-04
Duration: 1.5 hours	Date: 28.09.2022 Session (I/II/III): I	Max Marks: 20
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Essential Prime Implicants – These are those subcubes(groups) which cover atleast one minterm that can't be covered by any other prime implicant. **Essential prime implicants(EPI)** are those prime implicants which always appear in final solution.

Example:

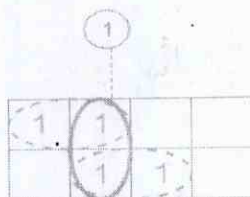


No. of Essential Prime Implicants = 2

Redundant Prime Implicants –

The prime implicants for which each of its minterm is covered by some essential prime implicant are **redundant prime implicants(RPI)**. This prime implicant never appears in final solution.

Example:

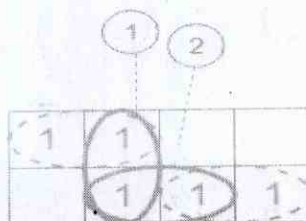


No. of Redundant Prime Implicants = 1

Selective Prime Implicants

The prime implicants for which are neither essential nor redundant prime implicants are called **selective prime implicants(SPI)**. These are also known as non-essential prime implicants. They may appear in some solution or may not appear in some solution.

Example:



No. of Selective Prime Implicants = 2



Solution of Question Paper
I Mid-Term Examination, Sept. -2022

Branch/Semester: III	Subject: DE	Subject Code: 3CAI3-04, 3CS3-04, 3DS3-04
Duration: 1.5 hours	Date: 28.09.2022 Session (I/II/III): I	Max Marks: 20
Submitted By: Shanu Tripathi, Dr. Vinay Kanungo, Dr. Niketa Sharma, Vinod Kataria		

Q.4 Describe 2's complement subtraction procedure and show the subtraction of $(10110)_2 - (11010)_2$.

Ans: The operation is carried out by means of the following steps:

- (i) At first, 2's complement of the subtrahend is found.
- (ii) Then it is added to the minuend.
- (iii) If the final carry over of the sum is 1, it is dropped and the result is positive.
- (iv) If there is no carry over, the two's complement of the sum will be the result and it is negative

2's complement of 11010 is $(00101 + 1)$ i.e. 00110. Hence

$$\begin{array}{r}
 \text{Minued -} \quad 10110 \\
 \text{2's complement of subtrahend -} \quad \underline{00110} \\
 \hline
 \text{Result of addition -} \quad 11100
 \end{array}$$

As there is no carry over, the result of subtraction is negative and is obtained by writing the 2's complement of 11100 i.e. $(00011 + 1)$ or 00100.

Hence the difference is -100 .

Q.5 Illustrate the following function in

(a) SOP minimal form using K-Maps:

$$F(A, B, C, D) = \sum_m (1, 2, 6, 7, 8, 13, 14, 15) + d(3, 5, 12)$$

(b) POS minimal form using K-Maps:

$$F(A, B, C, D) = \prod_M (1, 5, 6, 12, 13, 15) + d(2, 4)$$

Ans5(a) The SOP K-map for the given expression is:

		CD			
		00	01	11	10
AB	00		1	X	1
	01		X	1	1
	11	X	1	1	1
	10	1			

Therefore,



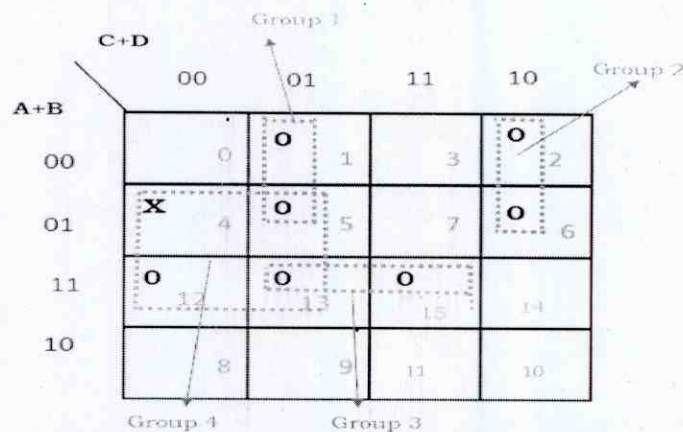
Solution of Question Paper

I Mid-Term Examination, Sept. -2022

Branch/Semester: III	Subject: DE	Subject Code: 3CAI3-04, 3CS3-04, 3DS3-04
Duration: 1.5 hours	Date: 28.09.2022	Session (I/II/III): I
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$$f = AC'D' + A'D + A'C + AB$$

Ans5(b)



K map with don't care condition for POS expression

First '0' is filled in the cells 1, 5, 6, 12, 13 and 14. The cells 2 and 4 are filled with 'X'.

The cells 2 and 4 are grouped along with the adjacent cells.

There are 4 groups.

1. In group 1 cells 1 and 5 are grouped and the common Maxterm is $A+C+D'$.
2. In group 2 cells 2 and 6 are grouped and the common Maxterm is $A+C'+D$.
3. In group 3 cells 13 and 15 are grouped and the common Maxterm is $A'+B'+D'$.
4. In group 4 cells 4, 5, 12 and 13 are grouped and the common Maxterm is $B'+C$.

So the minimized expression from K map is

$$f(A, B, C, D) = (A+C+D')(A+C'+D)(A'+B'+D')(B'+C)$$

Q.6 Determine the minimal sum of products for the Boolean expressions

$F(A, B, C, D) = \sum_m(2, 6, 8, 9, 10, 11, 14, 15)$ using the Quine-McCluskey method.

Ans.

1. min terms and their binary representations

Group A1 2 0010 →



Solution of Question Paper
I Mid-Term Examination, Sept. -2022

Branch/Semester: III	Subject: DE	Subject Code: 3CAI3-04, 3CS3-04, 3DS3-04
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	8	1000	→
	6	011	→
Group A2	9	1001	→
	10	1010	→
Group A3	11	1011	→
	14	1110	→
Group A4	15	1111	→

2. merging of min term

	2,6	0-10	→
Group B1	2,10	-010	→
(A1,A2)	8,9	100-	→
	8,10	10-0	→
	6,14	-110	→
Group B2	9,11	10-1	→
(A2,A3)	10,11	101-	→
	10,14	1-10	→
Group B3	11,15	1-11	→
(A3,A4)	14,15	111-	→

3. merging of min term pairs



Solution of Question Paper

I Mid-Term Examination, Sept. -2022

Branch/Semester: III	Subject: DE	Subject Code: 3CAI3-04, 3CS3-04, 3DS3-04
Duration: 1.5 hours	Date: 28.09.2022 Session (I/II/III): I	Max Marks: 20
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Group C1 2,6,10,14 --10 ✓
(B1,B2) 8,9,10,11 10-- ✓

Group C2 10,11,14,15 1-1- ✓
(B2,B3)

1. Prime implicant chart

PIs\Minterms	2	6	8	9	10	11	14	15	a,b,c,d
2,6,10,14	X	X			X		X		--10
8,9,10,11			X	X	X	X			10--
10,11,14,15					X	X	X	X	1-1-

Extracted essential prime implicants : --10,10--,1-1-

All extracted essential prime implicants : --10,10--,1-1-

Minimal QuineMcCluskey Expression = $cd' + ab' + ac$



**Swami Keshvanand Institute of Technology,
Management & Gramothan, Jaipur**
I Mid Term Examination, Sept.-2022

Semester:	III	Branch:	CSE/AI/DS/IT
Subject:	Data Structures and Algorithms	Subject Code:	3CS4-05/3CAI4-05/3AID4-05/3IT4-05
Time:	1.5 Hours	Maximum Marks:	20
Session (I/II/III): II			

PART A (short-answer type questions)

(All questions are compulsory)

(3*2=6)

Q.1 What is Complexity of an Algorithm?

Q.2 Differentiate between Static and Dynamic Memory Allocation.

Q.3 Differentiate between Linear Queue and Circular Queue.

PART B (Analytical/Problem solving questions)

(Attempt any 2 Questions)

(2*4=8)

Q.4 Write an algorithm for any two cases to insert an element in the linear linked list.

Q.5 Write an algorithm for insertion operation in circular queue using array.

Q.6 Explain Tower of Hanoi Problem for $n=3$ using recursion.

PART C (Descriptive/Analytical/Problem solving/Design questions)

(Attempt any 1 Question)

(1*6=6)

Q.7 Convert the following arithmetic expression into postfix expression:

$$A + (B * C - (D / E ^ F) * G) * H$$

Q.8 Write an algorithm for deleting last node from doubly linked list.



Question Paper Solution

Branch/Semester: III	Subject: DSA	Subject Code: 3CS4-05/3CA14-05/3AID4-05/3IT4-05
Duration: 1.5 hours	Date: 28.9.2022 Session (I/II/III): II	Max Marks: 20
Submitted By: Pawan Patidar, Ruchika Khandelwal, Mani Butwal, Nikhar Bhatnagar, Dr. Neha Janu, Rajesh Ranjan		

1. What is Complexity of an Algorithm?

Ans. Algorithmic complexity is a measure of how long an algorithm would take to complete given an input of size n . If an algorithm has to scale, it should compute the result within a finite and practical time bound even for large values of n . For this reason, complexity is calculated asymptotically as n approaches infinity.

Sometimes, there is more than one way to solve a problem. We need to learn how to compare the performance different algorithms and choose the best one to solve a particular problem. While analyzing an algorithm, we mostly consider time complexity and space complexity.

Time complexity of an algorithm quantifies the amount of time taken by an algorithm to run as a function of the length of the input. Similarly, Space complexity of an algorithm quantifies the amount of space or memory taken by an algorithm to run as a function of the length of the input.

Space Complexity: Space complexity is amount of space taken by an algorithm to complete the execution. Basically it is combination of two parts fixed part and variable part.

Time Complexity: Time complexity of an algorithm is a function $f(n)$ that describes the running time requirement for that algorithm.

2. Differentiate between Static and Dynamic Memory Allocation.

Ans.

S.No.	Static Memory Allocation	Dynamic Memory Allocation
1	In the static memory allocation, variables get allocated permanently, till the program executes or function call finishes.	In the Dynamic memory allocation, variables get allocated only if your program unit gets active.
2	Static Memory Allocation is done before program execution.	Dynamic Memory Allocation is done during program execution.
3	It uses <u>stack</u> for managing the static allocation of memory	It uses <u>heap</u> for managing the dynamic allocation of memory
4	It is less efficient	It is more efficient
5	In Static Memory Allocation, there is no memory re-usability	In Dynamic Memory Allocation, there is memory re-usability and memory can be freed when not required
6	In static memory allocation, once the memory is allocated, the memory size can not change.	In dynamic memory allocation, when memory is allocated the memory size can be changed.



Question Paper Solution

Branch/Semester: III	Subject: DSA	Subject Code: 3CS4-05/3CA14-05/3AID4-05/3IT4-05
Duration: 1.5 hours	Date: 28.9.2022 Session (I/II/III): II	Max Marks: 20
Submitted By: Pawan Patidar, Ruchika Khandelwal, Mani Butwal, Nikhar Bhatnagar, Dr. Neha Janu, Rajesh Ranjan		

- 7 In this memory allocation scheme, we cannot reuse the unused memory. This allows reusing the memory. The user can allocate more memory when required. Also, the user can release the memory when the user needs it.
- 8 In this memory allocation scheme, execution is faster than dynamic memory allocation. In this memory allocation scheme, execution is slower than static memory allocation.
- 9 In this memory is allocated at compile time. In this memory is allocated at run time.
- 10 In this allocated memory remains from start to end of the program. In this allocated memory can be released at any time during the program.
- 11 **Example:** This static memory allocation is generally used for array. **Example:** This dynamic memory allocation is generally used for linked list.

2. Differentiate between Linear Queue and Circular Queue.
Ans.

S.no.	Linear Queue	Circular Queue
1.	Arranges the data in a linear pattern.	Arranges the data in a circular order where the rear end is connected with the front end.
2.	The insertion and deletion operations are fixed i.e., done at the rear and front end respectively.	Insertion and deletion are not fixed and it can be done in any position.
3.	Linear queue requires more memory space.	It requires less memory space.
4.	In the case of a linear queue, the element added in the first position is going to be deleted in the first position. The order of operations performed on any element is fixed i.e., FIFO.	In the case of circular queue, the order of operations performed on an element may change.
5.	It is inefficient in comparison to a circular queue.	It is more efficient in comparison to linear queue.
6.	In a linear queue, we can easily fetch out the peek value.	In a circular queue, we cannot fetch out the peek value easily.



Question Paper Solution

Branch/Semester: III	Subject: DSA	Subject Code: 3CS4-05/3CAI4-05/3AID4-05/3IT4-05
Duration: 1.5 hours	Date: 28.9.2022 Session (I/II/III): II	Max Marks: 20
Submitted By: Pawan Patidar, Ruchika Khandelwal, Mani Butwal, Nikhar Bhatnagar, Dr. Neha Janu, Rajesh Ranjan		

7. Application- People standing for the bus.
Cars lined on a bridge.

Application- Computer-controlled traffic signal.
In CPU scheduling and memory management.

8. If there are 10 spaces then in the best case all the 10 spaces in the queue can be filled

If there are 10 spaces then in the best case 9 spaces can be filled at a time

4. Write an algorithm for any two cases to insert an element in the linear linked list.
Ans.

INSERTION AT BEGINNING

Step 1: IF AVAIL = NULL

Write OVERFLOW

Go to Step 7

[END OF IF]

Step 2: SET NEW_NODE = AVAIL

Step 3: SET AVAIL = AVAIL → NEXT

Step 4: SET NEW_NODE → DATA = VAL

Step 5: SET NEW_NODE → NEXT = HEAD

Step 6: SET HEAD = NEW_NODE

Step 7: EXIT

INSERTION IN SINGLY LINKED LIST AT THE END

Step 1: IF AVAIL = NULL Write OVERFLOW

Go to Step 1

[END OF IF]

Step 2: SET NEW_NODE = AVAIL

Step 3: SET AVAIL = AVAIL -> NEXT

Step 4: SET NEW_NODE -> DATA = VAL

Step 5: SET NEW_NODE -> NEXT = NULL

Step 6: SET PTR = HEAD

Step 7: Repeat Step 8 while PTR -> NEXT != NULL

Step 8: SET PTR = PTR -> NEXT

[END OF LOOP]

Step 9: SET PTR -> NEXT = NEW_NODE

Step 10: EXIT

5. Write an algorithm for insertion operation in queue using array.

Ans:

Algorithm Insert in queue

1. If (REAR = size) then //Queue is full
2. Write "Queue is full"
3. Exit



Question Paper Solution

Branch/Semester: III	Subject: DSA	Subject Code: 3CS4-05/3CA14-05/3AID4-05/3IT4-05
Duration: 1.5 hours	Date: 28.9.2022 Session (I/II/III): II	Max Marks:20
Submitted By: Pawan Patidar, Ruchika Khandelwal, Mani Butwal, Nikhar Bhatnagar, Dr. Neha Janu, Rajesh Ranjan		

4. Else

REAR = REAR + 1 // increment REAR

5. Que[REAR] = ITEM

6. If (FRONT = -1) then //Queue is empty

7. Set FRONT = 0

10. End if

Algorithm Delete in queue:

1. If (FRONT = -1) then

2. wwrite "Queue is empty"

3. Exit

4. Else

5. ITEM = Que [FRONT]

6. If (FRONT = REAR)

7. set REAR = -1

8. set FRONT = -1

9. Else

10. FRONT = FRONT + 1

11. End if

12. End if

13. Stop

6. Explain Tower of Hanoi Problem for $n=3$ using recursion.

It is a mathematical game or puzzle that consists of three rods with 'n' number of disks of different diameters. The objective of the game is to shift the entire stack of disks from one rod to another rod following these three rules :

1. Only one disk can be moved at a time.
2. Only the uppermost disk from one stack can be moved on to the top of another stack or an empty rod.
3. Larger disks cannot be placed on the top of smaller disks.

The minimal number of moves required to solve the Tower of Hanoi puzzle of n disks would be $(2^n) - 1$.

The logic behind solving the Tower of Hanoi for three disks :

Objective: To solve the Tower of Hanoi puzzle that contains three disks. The stack of disks has to be shifted from Rod 1 to Rod 3 by abiding to the set of rules that has been mentioned above.

Step 1 : The smallest green disk , the uppermost disk on the stack is shifted from rod 1 to rod 3.

Step 2 : Next the uppermost disk on rod 1 is the blue colored disk which is shifted to rod 2.

Step 3 : The smallest disk placed on rod 3 is shifted back on to the top of rod 2.

Step 4 : So now the largest red disk is allowed to be shifted from rod 1 to its destination rod 3.

Step 5 : Now the two disks on rod 2 has to be shifted to its destination rod 3 on top of the red disk , so first the smallest green disk on top of the blue rod is shifted to rod 1 .

Step 6 : Next the blue disk is permitted to be shifted to its destination rod 3 that will stacked on to the top of the red disk.

Step 7 : Finally , the smallest green colored rod is also shifted to rod 3 , which would now be the uppermost rod on the stack .

So the Tower of Hanoi for three disks has been solved!!

Algorithm

To write an algorithm for Tower of Hanoi, first we need to learn how to solve this problem with lesser amount



Question Paper Solution

Branch/Semester: III	Subject: DSA	Subject Code: 3CS4-05/3CA14-05/3AID4-05/3IT4-05
Duration: 1.5 hours	Date: 28.9.2022 Session (I/II/III): II	Max Marks:20
Submitted By: Pawan Patidar, Ruchika Khandelwal, Mani Butwal, Nikhar Bhatnagar, Dr. Neha Janu, Rajesh Ranjan		

of disks, say $\rightarrow 1$ or 2. We mark three towers with name, source, destination and aux (only to help moving the disks). If we have only one disk, then it can easily be moved from source to destination peg.

If we have 2 disks –

- First, we move the smaller (top) disk to aux peg.
- Then, we move the larger (bottom) disk to destination peg.
- And finally, we move the smaller disk from aux to destination peg.

The steps to follow are –

Step 1 – Move $n-1$ disks from source to aux

Step 2 – Move n^{th} disk from source to dest

Step 3 – Move $n-1$ disks from aux to dest

A recursive algorithm for Tower of Hanoi can be driven as follows –

START

Procedure Hanoi(disk, source, dest, aux)

IF disk == 1, THEN

move disk from source to dest

ELSE

Hanoi(disk - 1, source, aux, dest) // Step 1

move disk from source to dest // Step 2

Hanoi(disk - 1, aux, dest, source) // Step 3

END IF

END Procedure

STOP

7. Convert the following arithmetic expression into postfix expression:

$$X = A + (B * C - (D / E - F) * G) * H.$$

Ans: $X = A + (B * C - (D / E - F) * G) * H.$

Question Paper Solution

Branch/Semester: III	Subject: DSA	Subject Code: 3CS4-05/3CA14-05/3AID4-05/3IT4-05
Duration: 1.5 hours	Date: 28.9.2022 Session (I/II/III): II	Max Marks:20
Submitted By: Pawan Patidar, Ruchika Khandelwal, Mani Butwal, Nikhar Bhatnagar, Dr. Neha Janu, Rajesh Ranjan		

Symbol	Scanned	STACK	Postfix Expression	Description
1.		(		Start
2.	A	{	A	
3.	+	{+	A	
4.	(	{+(	A	
5.	B	{+(	AB	
6.	*	{+(*	AB	
7.	C	{+(*	ABC	
8.	-	{+(-	ABC*	'*' is at higher precedence than '-'
9.	(	{+(-(	ABC*	
10.	D	{+(-(	ABC*D	
11.	/	{+(-(/	ABC*D	
12.	E	{+(-(/	ABC*DE	
13.	^	{+(-(/^	ABC*DE	
14.	F	{+(-(/^	ABC*DEF	
15.	)	{+(-	ABC*DEF^/	Pop from top on Stack, that's why '^' Come first
16.	*	{+(-*	ABC*DEF^/	
17.	G	{+(-*	ABC*DEF^/G	
18.	)	{+(-	ABC*DEF^/G*-	Pop from top on Stack, that's why '^' Come first
19.	*	{+**	ABC*DEF^/G*-	
20.	H	{+**	ABC*DEF^/G*-H	
21.	)	Empty	ABC*DEF^/G*-H*+	END

8 Write an algorithm for deleting last node from doubly linked list.

Ans.

- STEP 1 : NOW IF HEAD = NULL (GO TO STEP 6)
- STEP 2 : ASSIGN EXTRA = HEAD
- STEP 3 : MAKE A LOOP WHILE EXTRA = NEXT != NULL
- STEP 4 : ASSIGN EXTRA = EXTRA -> NEXT (END THE WHILE LOOP)
- STEP 5 : THAN ASSIGN EXTRA -> PREVIOUS -> NEXT = NULL
- STEP 6 : THAN FREE EXTRA
- STEP 7 : EXIT



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I Mid Term Examination, Sept.-2022

Semester:	III	Branch:	CSE/CS(AI)/CS(DS)/IT
Subject:	Object Oriented Programming	Subject Code:	3CS4-06/3CAI4-06/3IT4-06
Time:	1.5 Hours	Maximum Marks:	20
Session (I/II/III): I			

PART A (short-answer type questions)

(All questions are compulsory)

(3*2=6)

- Q1. List all the characteristics of object oriented paradigm.
- Q2. What are the methods of defining the class member function? List all
- Q3. Define the role of public and private access specifiers within a class.

PART B (Analytical/Problem solving questions)

(Attempt any 2 Questions)

(2*4=8)

- Q4. Define classes and object? Explain the process of creating and destroying objects dynamically using "new" and "delete" operators with the help of example.
- Q5. What do you mean by function overloading? Explain with example.
- Q6. Write the difference between constructor and destructor.

PART C (Descriptive/Analytical/Problem solving/Design questions)

(Attempt any 1 Question)

(1*6=6)

- Q7. Write a C++ program to display student details (Name, Roll no, Branch, contact) using constructor (default, copy, parameterized)
- Q8. Develop a program which add two complex numbers and return an object from function.



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Ramnagar, Jagatpura, Jaipur-302017**

Solution of Question Paper

I Mid-Term Examination, Sept. -2022

Branch/Semester: CSE/CS(AI)/CS(DS)/IT/III	Subject: Object Oriented Programming	Subject Code: 3CS4-06/3CAI4-06/3IT4-06
Duration: 1.5 hours	Date: 22/09/2022 Session (I/II/III): I	Max Marks: 20
Submitted By: M.K.Beniwal, Kailash Soni, Ashish Pant, Garima Gupta, Priya Gupta, Shalini Singhal, Dolly Mittal		

PART A (short-answer type questions)

(All questions are compulsory)

(3*2=6)

Q1. List all the characteristics of object oriented paradigm.

[2]

Ans. Features of OOPs:

- Classes.
- Objects.
- Data Abstraction.
- Encapsulation.
- Inheritance.
- Polymorphism.

Q2. What are the methods of defining the class member function? List all

[2]

Ans.

Member Function:

A member function of a class is a function that has its definition or its prototype within the class definition like any other variable. If the member function is defined inside the class definition it can be defined directly, but if it is defined outside the class, then we have to use the scope resolution :: operator along with class name along with function name.

1) Member functions defined within the class

```
class className{
returnType MemberFunction(arguments)
{
//function body
}
....
};
```

2) Member functions defined outside the class

```
class className{
returnType MemberFunction(arguments);
....
}
```



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Ramnagar, Jagatpura, Jaipur-302017**

Solution of Question Paper
I Mid-Term Examination, Sept. -2022

Branch/Semester: CSE/CS(AI)/CS(DS)/IT/III	Subject: Object Oriented Programming	Subject Code: 3CS4-06/3CAI4-06/3IT4-06
Duration: 1.5 hours	Date: 22/09/2022 Session (I/II/III):I	Max Marks: 20
Submitted By: M.K.Beniwal, Kailash Soni, Ashish Pant, Garima Gupta, Priya Gupta, Shalini Singhal, Dolly Mittal		

```
};  
  
returnType className :: function_Name(arg1,arg2,.....argN)  
{  
    function body  
}
```

Q3. Define the role of public and private access specifiers within a class.

[2]

Ans.

1. Public: All the class members declared under the public specifier will be available to everyone. The data members and member functions declared as public can be accessed by other classes and functions too. The public members of a class can be accessed from anywhere in the program using the direct member access operator (.) with the object of that class.

2. Private: The class members declared as *private* can be accessed only by the member functions inside the class. They are not allowed to be accessed directly by any object or function outside the class. Only the member functions or the friend functions are allowed to access the private data members of a class.

PART B (Analytical/Problem solving questions)

(Attempt any 2 Questions)

(2*4=8)

Q4. Define classes and object? Explain the process of creating and destroying objects dynamically using “new” and “delete” operators with the help of example.

[4]

Ans.

Class: Class is a user defined data type, which holds its own data members and member functions, which can be accessed and used by creating instance of that class.

Object: Objects are instances of class, which holds the data variables declared in class and the member functions work on these class objects.

```
#include <iostream.h>
```

```
int main ()  
{  
    // Pointer initialization to null
```



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```
int* p = NULL;
```

```
// Request memory for the variable
```

```
// using new operator
```

```
p = new(nothrow) int;
```

```
if (!p)
```

```
    cout << "allocation of memory failed\n";
```

```
else
```

```
{
```

```
    // Store value at allocated address
```

```
    *p = 29;
```

```
    cout << "Value of p: " << *p << endl;
```

```
}
```

```
// Request block of memory
```

```
// using new operator
```

```
float *r = new float(75.25);
```

```
cout << "Value of r: " << *r << endl;
```

```
// Request block of memory of size n
```

```
int n = 5;
```

```
int *q = new(nothrow) int[n];
```

```
if (!q)
```

```
    cout << "allocation of memory failed\n";
```

```
else
```

```
{
```

```
    for (int i = 0; i < n; i++)
```

```
        q[i] = i+1;
```

```
    cout << "Value store in block of memory: ";
```

```
    for (int i = 0; i < n; i++)
```

```
        cout << q[i] << " ";
```

```
}
```

```
// freed the allocated memory
```



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```
delete p;
delete r;

// freed the block of allocated memory
delete[] q;

return 0;
}
```

Q5. What do you mean by function overloading? Explain with example. [4]

Ans. Function overloading is a C++ programming feature that allows us to have more than one function having same name but different parameter list, when I say parameter list, it means the data type and sequence of the parameters, for example the parameters list of a function myfuncn(int a, float b) is (int, float) which is different from the function myfuncn(float a, int b) parameter list (float, int). Function overloading is a compile-time polymorphism.

// Program to compute absolute value: Works for both int and float

```
#include <iostream.h>
```

```
    // function with float type parameter
```

```
float absolute(float var){
```

```
    if (var < 0.0)
```

```
        var = -var;
```

```
    return var;
```

```
}
```

```
// function with int type parameter
```

```
int absolute(int var) {
```

```
    if (var < 0)
```

```
        var = -var;
```



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```
return var;
}
int main() {
    // call function with int type parameter
    cout << "Absolute value of -5 = " << absolute(-5) << endl;
    // call function with float type parameter
    cout << "Absolute value of 5.5 = " << absolute(5.5f) << endl;
    return 0;
}
```

Q6. Write the difference between constructor and destructor.

[4]

S.no.	Parameter of Difference	Constructor	Destructor
1.	Initialization	Initialized with the same as that of the class name. i.e., Example: if the class name is maxim, the constructor for the same will be maxim()	Also initialized with the same name as that of the class name, but with a tilde (~) sign before the class name. Example: if the class name is maxim, the constructor for the same will be ~maxim()
2.	Objective	Creates the class objects. Or, When a constructor is called, it allocates the	Used for destroying the objects. i.e., when a destructor is called, memory is released.



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		memory.	
3.	Parameters/arguments	Accepts the arguments. Example: maxim(a,b,c)	Does not contain any arguments. Example : maxim()
4.	Overloading	Posses Overloading property. Multiple constructors exist in a class at a time.	Does not possess the property of overloading; (Also there is no need of destructor overloading). Only one destructor can exist in one class at a time.
5.	Invoking	As soon as the object is created, a constructor is called by default. Also, they are called in successive order; in the order they are designed.	When the concerned block is encountered or if the program is terminated, Destructor is called. They are called in the reverse order of the design.
6.	Types	Constructor can be of types: Default (with no arguments) and parameterized (with arguments).	Destructor can never be parameterized (as mentioned earlier).
7.	Copying the values	The copy constructor lets us	There is no like functionality in



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		declare and initialize the object from some other object.	the Destructor.
8.	Syntax	<pre>Class_name(a,b,c) { //body of the constructor }</pre>	<pre>~Class_name() { //body of the destructor }</pre>

PART C (Descriptive/Analytical/Problem solving/Design questions)

(Attempt any 1 Question)

(1*6=6)

Q7. Write a C++ program to display student details(Name,Roll no, Branch, contact) using constructor (default, copy,parameterized) [6]

Ans.

```
#include<iostream>
using namespace std;
class stu
{
private:
char name[20],add[20];
int roll,zip;
```



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public:

stu ();//Constructor

~stu();//Destructor

stu (int r,)

{

roll =r;

}

stu (stu &s)

{

}

void read();

void disp();

};

stu :: stu()

{

cout<<"\nThis is Student Details constructor called....."<<endl;

}

void stu :: read()

{

cout<<"\nEnter the student Name :: ";



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```
cin>>name;

cout<<"\nEnter the student roll no :: ";

cin>>roll;

cout<<"\nEnter the student address :: ";

cin>>add;

cout<<"\nEnter the Zipcode :: ";

cin>>zip;

}

void stu :: disp()

{

cout<<"\nThe Entered Student Details are shown below ::----- \n";

cout<<"\nStudent Name :: "<<name<<endl;

cout<<"\nRoll no is :: "<<roll<<endl;

cout<<"\nAddress is :: "<<add<<endl;

cout<<"\nZipcode is :: "<<zip;

}

stu :: ~stu()

{

cout<<"\n\nStudent Detail is Closed.....\n";

}
```



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```
int main()
{
    stu s;
    s.read ();
    s.disp ();
    stu s1(92356);
    s1.disp();
    stu s2(s1);
    s2.disp();
    return 0;
}}
```

Q8. Develop a program which add two complex numbers and return an object from function.

[6]

Ans.

```
/* C++ program to Add two Complex number passing objects to function */
#include<iostream>
using namespace std;
class complex
{
    int re,im;
    public:
```



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```
void get()
{
    cout<<"\nEnter Real Part :: ";
    cin>>re;
    cout<<"\nEnter Imag. Part :: ";
    cin>>im;
}

void disp()
{
    cout<<re<<"+"<<im<<"i"<<"\n";
}

complex sum(complex,complex);
};

complex complex::sum(complex c1,complex c2)
{
    complex c3;
    c3.re=c1.re+c2.re;
    c3.im=c1.im+c2.im;
    return c3;
}

int main()
{
```



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Duration: 1.5 hours	Date: 22/09/2022 Session (I/II/III):I	Max Marks: 20
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```
complex c1,c2,c3;

cout<<"Enter 1st complex no.: \n";

c1.get();

cout<<"\nEnter 2nd complex no.: \n";

c2.get();

cout<<"\nThe 1st complex no. is :: ";

c1.disp();

cout<<"\nThe 2nd complex no. is :: ";

c2.disp();

c3=c3.sum(c1,c2);

cout<<"\nThe Sum of two complex no.s are :: ";

c3.disp();

return 0;

}
```



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I Mid Term Examination, Sept.-2022**

Semester:	III	Branch:	CS/DS/AI
Subject:	Software Engineering	Subject Code:	3CS4-07/ 3AID4-07
Time:	1.5 Hours	Maximum Marks:	20

Session (I/II/III):II

PART A (short-answer type questions)

(All questions are compulsory)

(3*2=6)

- Q1. Define Software Engineering.
Q2. Differentiate generic software and customized software.
Q3. List various types of risks in software development.

PART B (Analytical/Problem solving questions)

(Attempt any 2 Questions)

(2*4=8)

- Q4. Describe characteristics of a good software.
Q5. Differentiate Validation and Verification.
Q6. Compare Line of Code and Function Point estimation.

PART C (Descriptive/Analytical/Problem solving/Design questions)

(Attempt any 1 Question)

(1*6=6)

- Q7. Explain Waterfall model in detail with a neat diagram.
Q8. A project was estimated to be 400 KLOC. Calculate the effort and development time for each of the three model (organic, semi-detached & embedded) in Basic COCOMO Model.

Mode	a	b	c	d
Organic	2.4	1.05	2.5	0.38
Semi Detached	3.0	1.12	2.5	0.35
Embedded	3.6	1.20	2.5	0.32



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I Mid Term Examination, Sept.-2022**

Semester:	III	Branch:	CS/DS/AI
Subject:	Software Engineering	Subject Code:	3CS4-07/ 3AID4-07
Time:	1.5 Hours	Maximum Marks:	20

Session (I/II/III):II

PART A (short-answer type questions)

(All questions are compulsory)

(3*2=6)

- Q1. Define Software Engineering.
Q2. Differentiate generic software and customized software.
Q3. List various types of risks in software development.

PART B (Analytical/Problem solving questions)

(Attempt any 2 Questions)

(2*4=8)

- Q4. Describe characteristics of a good software.
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PART C (Descriptive/Analytical/Problem solving/Design questions)

(Attempt any 1 Question)

(1*6=6)

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Solution of Question Paper
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Branch/Semester: CS/DS/AI/III	Subject: Software Engineering	Subject Code: 3CS4-07// 3AID4-07
Duration: 1.5 hours	Date: 30/9/2022 Session (I/II/III): II	Max Marks:20
Submitted By: ALLISA GOYAL		

PART A (short-answer type questions)

Q1. Define Software Engineering.

ANS: The term software engineering is the product of two words, software, and engineering.

The software is a collection of integrated programs. Software subsists of carefully-organized instructions and code written by developers on any of various particular computer languages.

Engineering is the application of scientific and practical knowledge to invent, design, build, maintain, and improve frameworks, processes, etc.

Software Engineering is an engineering branch related to the evolution of software product using well-defined scientific principles, techniques, and procedures. The result of software engineering is an effective and reliable software product.

Q2. Differentiate generic software and customized software.

ANS:

S.No.	Generic software product development	Custom software development
1.	The generic software development is done for developing general purpose software.	Customer software development is done to develop a software product as per the needs of particular customer.
2.	In this development process, the software developers have to depict the end-users specifications.	In this development process, the end-user requirements can be aggregated by communicating by them.



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3.	From designing and marketing perspective, this type of development is very difficult.	This development does not require marketing, because it is developed for appropriate group of users.
4.	Large number of users may be using this kind of software.	This type of software is used by limited number of users.
5.	Quality of the product is not a preference for generic software.	Quality is the main criterion in customer software product. Best quality of the product is focused for customer or company.
6.	Development team controls the process of generic software development.	Customer determines the process of software development in this type of product.
7.	Generally the software developed is economical. There may be some hidden costs such as installation and implementation cost.	Software product is of high cost as the particular product for customer is developed.
8.	Example of generic software product development is Word-editing software.	Inventory control and management system are examples of customer software development.

Q3. List various types of risks in software development.

ANS: 1. Schedule Risk:

Schedule related risks refer to time related risks or project delivery related planning risks. The wrong schedule affects the project development and delivery.



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2. Budget Risk:

Budget related risks refer to the monetary risks mainly it occurs due to budget overruns. Always the financial aspect for the project should be managed as per decided but if financial aspect of project mismanaged then there budget concerns will arise by giving rise to budget risks.

3. Operational Risks:

Operational risk refers to the procedural risks means these are the risks which happen in day-to-day operational activities during project development due to improper process implementation or some external operational risks.

4. Technical Risks :

Technical risks refers to the functional risk or performance risk which means this technical risk mainly associated with functionality of product or performance part of the software product.

5. Programmatic Risks:

Programmatic risks refer to the external risk or other unavoidable risks. These are the external risks which are unavoidable in nature. These risks come from outside and it is out of control of programs.

PART B (Analytical/Problem solving questions)

Q4. Describe characteristics of good software.

ANS: A software product can be judged by what it offers and how well it can be used. This software must satisfy on the following grounds:

- Operational
- Transitional
- Maintenance

Well-engineered and crafted software is expected to have the following characteristics:

Operational

This tells us how well software works in operations. It can be measured on:

- Budget
- Usability
- Efficiency
- Correctness
- Functionality
- Dependability



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Submitted By: ALLISA GOYAL		

- Security
- Safety

Transitional

This aspect is important when the software is moved from one platform to another:

- Portability
- Interoperability
- Reusability
- Adaptability

Maintenance

This aspect briefs about how well a software has the capabilities to maintain itself in the ever-changing environment:

- Modularity
- Maintainability
- Flexibility
- Scalability

Q5. Differentiate Validation and Verification.

ANS:

Verification	Validation
• The verifying process includes checking documents, design, code, and program	• It is a dynamic mechanism of testing and validating the actual product
• It does not involve executing the code	• It always involves executing the code
• Verification uses methods like reviews, walkthroughs, inspections, and desk- checking etc.	• It uses methods like Black Box Testing, White Box Testing, and non-functional testing
• Whether the software conforms to specification is checked	• It checks whether the software meets the requirements and expectations of a customer
• It finds bugs early in the development cycle	• It can find bugs that the verification process can not catch
• Target is application and software architecture,	• Target is an actual product



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specification, complete design, high level, and database design etc.	
• QA team does verification and make sure that the software is as per the requirement in the SRS document.	• With the involvement of testing team validation is executed on software code.
• It comes before validation	• It comes after verification

Q6. Compare Line of Code and Function Point estimation

ANS:

• Function Point (FP)	• Line of Code (LOC)
• Function Point metric is specification-based.	• LOC metric is based on analogy.
• Function Point metric is language independent.	• LOC metric is dependent on language.
• Function Point metric is user-oriented.	• LOC metric is design-oriented.
• Function Point metric is extendible to Line of Code.	• It is changeable to FP (i.e, backfiring)
• Function Point is used for data processing systems	• LOC is used for calculating the size of the computer program
• Function Point can be used to portray the project time	• LOC is used for calculating and comparing the productivity of programmers.



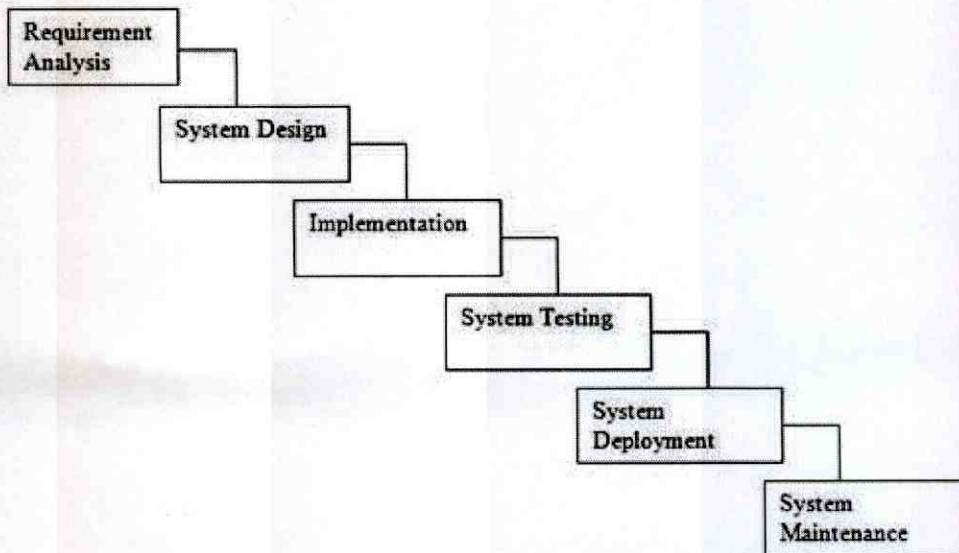
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PART C (Descriptive/Analytical/Problem solving/Design questions)

Q7. Explain Waterfall model in detail with a neat diagram.

ANS Waterfall model is an example of a Sequential model. In this model, the software development activity is divided into different phases and each phase consists of a series of tasks and has different objectives. Waterfall model is the pioneer of the SDLC processes. In fact, it was the first model which was widely used in the software industry. It is divided into phases and the output of one phase becomes the input of the next phase. It is mandatory for a phase to be completed before the next phase starts. In short, there is no overlapping in the Waterfall model. In waterfall, the development of one phase starts only when the previous phase is complete. Because of this nature, each phase of the waterfall model is quite precise and well-defined. Since the phases fall from a higher level to a lower level, like a waterfall, It's named the waterfall model.



The activities involved in different phases are as follows:

S.No	Phase	Activities Performed	Deliverables
1	Requirement Analysis	1. Capture all the requirements. 2. Do brainstorming and walkthrough to understand the requirements. 3. Do the requirements feasibility test to ensure that the requirements are testable or not.	RUD (Requirements Understanding Document)



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2	System Design	1. As per the requirements, create the design 2. Capture the hardware / software requirements. 3. Document the designs	HLD (High Level Design document) LLD (Low level design document)
3	Implementation	1. As per the design create the programmes / code 2. Integrate the codes for the next phase. 3. Unit testing of the code	Programs Unit test cases and results
4	System Testing	1. Integrate the unit tested code and test it to make sure if it works as expected. 2. Perform all the testing activities (Functional and non functional) to make sure that the system meets the requirements. 3. In case of any anomaly, report it. 4. Track your progress on testing through tools like traceability metrics, ALM 5. Report your testing activities.	Test cases Test reports Defect reports Updated matrices.
5	System Deployment	1. Make sure that the environment is up 2. Make sure that there are no sev 1 defects open. 3. Make sure that the test exit criteria are met. 4. Deploy the application in the respective environment. 5. Perform a sanity check in the environment after the application is deployed to ensure the application does not break.	User Manual Environment definition / specification
6	System maintenance	1. Make sure that the application is up and running in the respective environment. 2. Incase user encounters and defect, make sure to note and fix the issues faced. 3. Incase any issue is fixed; the updated code is deployed in the environment. 4. The application is always enhanced to incorporate more features, update the environment with the latest features	User Manual List of production tickets List of new features implemented.

The advantages of using the Waterfall model are as follows:

- Simple and easy to understand and use.



Solution of Question Paper
I Mid-Term Examination, Sept. -2022

Branch/Semester: CS/DS/AI/III	Subject: Software Engineering	Subject Code: 3CS4-07// 3AID4-07
Duration: 1.5 hours	Date: 30/9/2022 Session (I/II/III): II	Max Marks:20
Submitted By: ALLISA GOYAL		

- For smaller projects, the waterfall model works well and yields the appropriate results.
- Since the phases are rigid and precise, one phase is done one at a time, it is easy to maintain.
- The entry and exit criteria are well defined, so it is easy and systematic to proceed with quality.
- Results are well documented.
-

Disadvantages of using Waterfall model:

- Cannot adopt the changes in requirements
- It becomes very difficult to move back to the phase. For example, if the application has now moved to the testing stage and there is a change in requirement, It becomes difficult to go back and change it.
- Delivery of the final product is late as there is no prototype that is demonstrated immediately.
- For bigger and more complex projects, this model is not good as the risk factor is higher.
- Not suitable for projects where requirements are changed frequently.
- Does not work for long and ongoing projects.
- Since the testing is done at a later stage, it does not allow identifying the challenges and risks in the earlier phase so the risk mitigation strategy is difficult to prepare.
-

Q8. A project was estimated to be 400 KLOC. Calculate the effort and development time for each of the three model (organic, semi-detached & embedded) in Basic COCOMO Model.

Mode	a	b	c	d
Organic	2.4	1.05	2.5	0.38
Semi Detached	3.0	1.12	2.5	0.35
Embedded	3.6	1.20	2.5	0.32

ANS: The basic COCOMO equation takes the form:

$$\text{Effort} = a_1 * (\text{KLOC}) a_2 \text{ PM}$$

$$\text{Tdev} = b_1 * (\text{efforts}) b_2 \text{ Months}$$

$$\text{Estimated Size of project} = 400 \text{ KLOC}$$

(i)Organic Mode

$$E = 2.4 * (400)1.05 = 1295.31 \text{ PM}$$



Solution of Question Paper

I Mid-Term Examination, Sept. -2022

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$$D = 2.5 * (1295.31)0.38=38.07 \text{ PM}$$

(ii) Semidetached Mode

$$E = 3.0 * (400)1.12=2462.79 \text{ PM}$$

$$D = 2.5 * (2462.79)0.35=38.45 \text{ PM}$$

(iii) Embedded Mode

$$E = 3.6 * (400)1.20 = 4772.81 \text{ PM}$$

$$D = 2.5 * (4772.8)0.32 = 38 \text{ PM}$$



**Swami Keshvanand Institute of Technology,
Management & Gramothan, Jaipur
I Mid Term Examination, Sept.-2022**

Semester:	III	Branch:	IT
Subject:	Digital Electronics	Subject Code:	3IT3-04
Time:	1.5 Hours	Maximum Marks:	20
Session :			

PART A (short-answer type questions)

(All questions are compulsory)

(3*2=6)

- Q.1 Find the value of X in $(123)_{10} + (X)_{10} = (BA)_{16}$.
- Q.2 Why Gray Code is called Unit distance code? Convert $(10011001)_2$ into Gray Code.
- Q.3 Find radix r of $(23)_r + (12)_r = (101)_r$.

PART B (Analytical/Problem solving questions)

(Attempt any 2 Questions)

(2*4=8)

- Q.4 Reduce the given functions using Boolean algebra and realize using NAND Gates.
- $A [B + C (A B + A C)]$
- Q.5 Reduce the given functions using Boolean algebra and realize using Basic Gates.
- $(A + B C + C D) + B C$
- Q.6 Perform $(429)_{10} - (546)_{10}$ using BCD subtraction.

PART C (Descriptive/Analytical/Problem solving/Design questions)

(Attempt any 1 Question)

(1*6=6)

- Q.7 Minimize the given Boolean Function using K-Map.
- $\sum_m (1, 4, 8, 12, 13, 15) + d(3, 14)$ using K-Map.
- Q.8 Realize the given function Using K Map.
- $F(A, B, C, D) = \sum_m (0, 1, 2, 3, 7, 8, 9, 10, 11, 12, 13, 14)$



Solution of Question Paper
I Mid-Term Examination, Sept. -2022

Branch/Semester: IT/III	Subject: DE	Subject Code: 3IT3-04
Duration: 1.5 hours	Date: 28/09/22	Session (I/II/III): I
Submitted By: Dr. Sunita Gupta		
Max Marks: 20		

Q.1. $(123)_{10} + (X)_{10} = (BA)_{16}$ PART A

Convert Hexadecimal into decimal.

$$123 + X = B \times 16^1 + A \times 16^0 \Rightarrow 123 + X = 176 + 10 \quad (A=10, B=11)$$

$$\Rightarrow X = 176 - 123 = \boxed{53}$$

→ Each succeeding Code changes by only 1-bit from the Preceding Code. So Gray Code is called Unit distance Code.

Q.2. $(10011001)_2$
Gray Code is $\boxed{11010101}$

Q.3. $(23)_r + (12)_r = (101)_r$

Converting the radix r into base 10 we get

$$(2 \times r^1 + 3 \times r^0) + (1 \times r^1 + 2 \times r^0) = (1 \times r^2 + 0 \times r^1 + 1 \times r^0)$$

$$\Rightarrow 2r + 3 + r + 2 = r^2 + 1$$

$$\Rightarrow 3r + 5 = r^2 + 1$$

$$\Rightarrow (r-4)(r+1) = 0 \Rightarrow r = 4 \text{ \& } r = -1$$

$r = 4$ as base cannot be negative.

PART - B

Q.4. Given function is

$$f = A[B + C(AB + AC)]$$

$$\Rightarrow f = A[B + ABC + AC]$$

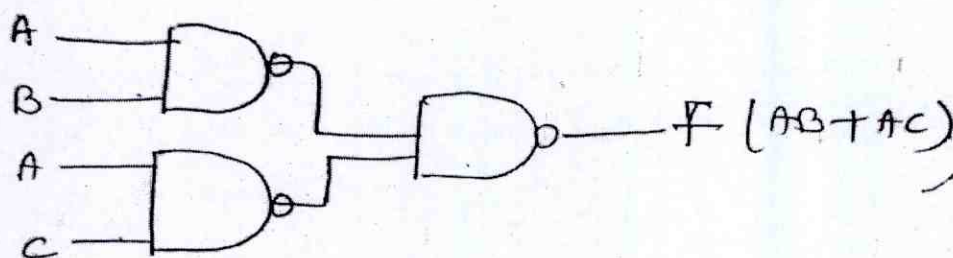
Solution of Question Paper
I Mid-Term Examination, Sept. -2022

Branch/Semester: IT/III	Subject: DE	Subject Code: 3IT3-04
Duration: 1.5 hours	Date: 28/09/22	Session (I/II/III): I
Submitted By: Dr. Sunita Gupta		Max Marks: 20

$$\Rightarrow f = AB + ABC + AC$$

$$\Rightarrow f = AB + AC(B+1)$$

$$\Rightarrow f = AB + AC$$



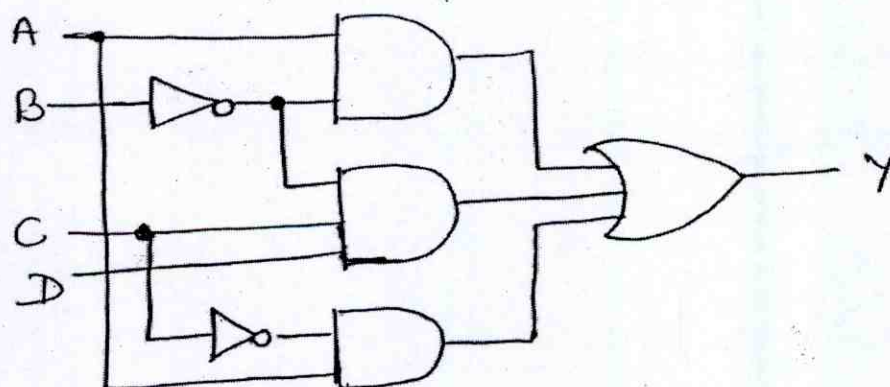
Q.5
$$\overline{(A + BC + CD)} + BC$$

$$= \overline{(A + BC + CD)} \cdot \overline{BC}$$

$$= (A + BC + CD) \cdot (\overline{B} + \overline{C})$$

$$= A\overline{B} + B\overline{B}C + \overline{B}CD + A\overline{C} + B\overline{C} + C\overline{C}D$$

$$= A\overline{B} + \overline{B}CD + A\overline{C}$$





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Submitted By: Dr. Sunita Gupta		

Q.6 429 - 546 using BCD.

$$\begin{array}{r} 0100 \quad 0010 \quad 1001 \rightarrow \text{minuend} \\ - 0101 \quad 0100 \quad 0110 \rightarrow \text{subtrahend} \\ \hline \end{array}$$

↓
Taking 1's Complement of it
and adding to minuend

$$\begin{array}{r} 0100 \quad 0010 \quad 1001 \\ + 0101 \quad 0100 \quad 0110 \\ \hline 1110 \quad 1110 \quad 0010 \\ \downarrow \quad \downarrow \quad \downarrow \\ C_n \quad C_n \quad C_n \end{array}$$

As BAC = 0,

$$\begin{array}{r} 0001 \quad 0001 \quad 1101 \\ + 0000 \quad 0000 \quad 0101 \\ \hline 0001 \quad 0001 \quad 0111 \end{array} \rightarrow X.$$

$= -(117)_{10}$ result

Q.7 $F(A, B, C, D) = \sum m(1, 4, 8, 12, 13, 15) + d(3, 14)$

4-variable K-map will be used.

In minterm, a product term will contain all the literals.
in SOP.

In maxterm, a sum term will contain all the literals
in POS.

Solution of Question Paper
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Submitted By: Dr. Sunita Gupta		Max Marks: 20

CD \ AB	00	01	11	10
00	0	1	1	1
01	1	0	1	0
11	X	0	1	0
10	0	0	X	0

$$Y = AB + B\bar{C}\bar{D} + A\bar{C}\bar{D} + \bar{A}\bar{B}D$$

Q. 8 $f(A, B, C, D) = \sum m(0, 1, 2, 3, 7, 8, 9, 10, 11, 12, 13, 14)$

CD \ AB	00	01	11	10
00	1	0	1	1
01	1	0	1	1
11	1	1	0	1
10	1	0	0	1

$$f = \bar{B} + A\bar{C} + \bar{A}CD$$

Standard SOP - is the sum of all minterms for which the function has the value 1.



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I Mid Term Examination, Sept.-2022

Semester:	III	Branch:	IT
Subject:	SOFTWARE ENGINEERING	Subject Code:	3IT4-07
Time:	1.5 Hours	Maximum Marks:	20

Session (I/II/III): II

PART A (short-answer type questions)

(All questions are compulsory)

(3*2=6)

Q.1 Describe the key elements of Software Engineering and write phases of SDLC.

Q.2 Describe the various phases of risk management in brief.

Q.3 What is SRS? Mention the qualities of a good SRS.

PART B (Analytical/Problem solving questions)

(Attempt any 2 Questions)

(2*4=8)

Q.4 Explain scope and importance of SPM (Software Project Management). Write the method of software size estimation technique.

Q.5 What are the characteristics of good software? Explain Spiral model in detail. Name any two systems for which this model is suitable.

Q.6 Describe characteristics of iterative model. Explain verification v/s validation and the different techniques of verification.

PART C (Descriptive/Analytical/Problem solving/Design questions)

(Attempt any 1 Question)

(1*6=6)

Q.7 Explain Agile Model in detail. Also differentiates the characteristics between Incremental Model and RAD Model.

Q.8 Suppose that a project was estimated to be 800000 LOC. Calculate effort, development time, human resources and productivity for each of 3 modes of development for COCOMO Basic Model.

Software Product Type	a	b	c	d
Organic	2.4	1.05	2.5	0.38
Semi-detached	3.0	1.12	2.5	0.35
Embedded	3.6	1.20	2.5	0.32



Solution of Question Paper
I Mid-Term Examination, Sept. -2022

Branch/Semester:III....	Subject:SOFTWARE ENGINEERING...	Subject Code: 3IT4-07
Duration: 1.5 hours	Date:-30/09/2022 Session:- II	Max Marks:- 20
Submitted By:- Ms. Sanju Choudhary		

PART A (short-answer type questions)
(All questions are compulsory)

(3*2=6)

Q.1 Describe the key elements of Software Engineering and write phases of SDLC.

Ans.:-

Three Key Elements in SE

Process: life-cycle model used, project management and assessment, quality assurance, etc.

Method: Approaches for solving a particular problem.

The “how to’s” for doing some specific task. E.g. object-oriented design; black-box testing; prototypes for requirements analysis.

Tools: Software that supports methods and/or processes. E.g. CASE: Computer-aided Software Engineering, Test environments, 4GLs, Design tools, etc.

System Development Life Cycle (SDLC) is a series of six main phases to create a hardware system “A system is a set of interacting or interdependent components forming an integrated whole” it’s a term that can be used in different industries, therefore Software Development Life Cycle is a limited term that explains the phases of creating a software component that integrates with other software components to create the whole system. Some more specific takes on SDLC include:

1. System Planning

The Planning phase is the most crucial step in creating a successful system, during this phase you decide exactly what you want to do and the problems you’re trying to solve.

2- System Analysis

The end-user’s requirements should be determined and documented, what their expectations are for the system, and how it will perform.

3- System Design

The design phase comes after a good understanding of customer’s requirements; this phase defines the elements of a system, the components, the security level, modules, architecture and the different interfaces and type of data that goes through the system.

4- Implementation and Deployment

This phase comes after a complete understanding of system requirements and specifications; it’s the actual construction process after having a complete and illustrated design for the requested system.

5- System Testing and Integration

Bringing different components and subsystems together to create the whole integrated system, and then introducing the system to different inputs to obtain and analyze its outputs and behaviour and the way it functions. Testing is becoming more and more important to ensure customer’s satisfaction, and it requires no knowledge in coding, hardware configuration or design.

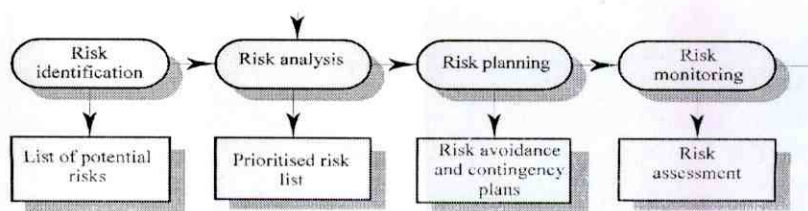
6- System Maintenance

In this phase, periodic maintenance for the system will be carried out to make sure that the system won’t become obsolete, this will include replacing the old hardware and continuously evaluating system’s performance, it also includes providing latest updates for certain components to make sure it meets the right standards and the latest technologies to face current security threats.

Q.2 Describe the various phases of risk management in brief.

Ans.:-

- **Risk identification – Identify project, product and business risks**
- **Risk analysis – Assess the likelihood and consequences of risks**
- **Risk planning – Draw up plans to avoid/minimise risk effects**
- **Risk monitoring – Monitor the risks throughout the project**





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Submitted By:- Ms. Sanju Choudhary		

Q.3 What is SRS? Mention the qualities of a good SRS.

Ans.:- A software requirements specification (SRS) is a document that captures complete description about how the system is expected to perform. It is usually signed off at the end of requirements engineering phase.
The sequential phases in Waterfall model are –

- Correctness:**
User review is used to ensure the correctness of requirements stated in the SRS. SRS is said to be correct if it covers all the requirements that are actually expected from the system.
- Completeness:**
Completeness of SRS indicates every sense of completion including the numbering of all the pages, resolving the to be determined parts to as much extent as possible as well as covering all the functional and non-functional requirements properly.
- Consistency:**
Requirements in SRS are said to be consistent if there are no conflicts between any set of requirements. Examples of conflict include differences in terminologies used at separate places, logical conflicts like time period of report generation, etc.
- Unambiguousness:**
A SRS is said to be unambiguous if all the requirements stated have only 1 interpretation. Some of the ways to prevent unambiguousness include the use of modelling techniques like ER diagrams, proper reviews and buddy checks, etc.
- Ranking for importance and stability:** There should a criterion to classify the requirements as less or more important or more specifically as desirable or essential. An identifier mark can be used with every requirement to indicate its rank or stability.
- Modifiability:**
SRS should be made as modifiable as possible and should be capable of easily accepting changes to the system to some extent. Modifications should be properly indexed and cross-referenced.
- Verifiability:**
A SRS is verifiable if there exists a specific technique to quantifiably measure the extent to which every requirement is met by the system. For example, a requirement stating that the system must be user-friendly is not verifiable and listing such requirements should be avoided.
- Traceability:**
One should be able to trace a requirement to design component and then to code segment in the program. Similarly, one should be able to trace a requirement to the corresponding test cases.
- Design Independence:**
There should be an option to choose from multiple design alternatives for the final system. More specifically, the SRS should not include any implementation details.
- Testability:**
A SRS should be written in such a way that it is easy to generate test cases and test plans from the document.

PART B (Analytical/Problem solving questions)
(Attempt any 2 Questions)

(2*4=8)

Q.4 Explain scope and importance of SPM (Software Project Management). Write the method of software size estimation technique.

Ans.:- SPM (Software Project Management): Project scope management is a process that helps in determining and documenting the list of all the project goals, tasks, deliverables, deadlines, and budgets as a part of the planning process. In project management, it is common for a big project to have modifications along the way. With the scope in the project management defined right in the beginning, it becomes much easier for project teams to manage and make the required changes.

Software Project Management (SPM) is a proper way of planning and leading software projects. It is a part of project management in which software projects are planned, implemented, monitored and controlled.

Need of Software Project Management:

Software is an non-physical product. Software development is a new stream in business and there is very little experience in building software products. Most of the software products are made to fit client's requirements. The most important is that the basic technology changes and advances so frequently and rapidly that experience of one product may not be applied to the other one. Such type of business and environmental constraints increase risk in software development hence it is essential to manage software projects efficiently.

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It is necessary for an organization to deliver quality product, keeping the cost within client's budget constrain and



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Submitted By:- Ms. Sanju Choudhary		

deliver the project as per scheduled. Hence in order, software project management is necessary to incorporate user requirements along with budget and time constraints.

Lines of Code (LOC):

LOC count the total number of lines of source code in a project. The units of LOC are:

- KLOC- Thousand lines of code
- NLOC- Non comment lines of code
- KDSI- Thousands of delivered source instruction

Function Point Analysis: In this method, the number and type of functions supported by the software are utilized to find FPC(function point count). The steps in function point analysis are:

- Count the number of functions of each proposed type.
- Compute the Unadjusted Function Points (UFP).
- Find Total Degree of Influence (TDI).
- Compute Value Adjustment Factor (VAF).
- Find the Function Point Count (FPC).

Both FP and LOC are units of measurement for software size. The size of a software that is subject to development is required in order to come up with accurate estimates of effort, cost and duration of a software project. Most parametric estimation models such as COCOMO accept size expressed in either FP or LOC as input.

Most of developer prefers functional size of software expressed as FP for one very important reason: Size expressed using the FP metric stay constant regardless of which programming language or languages are used.

Since the industry has more than 700 programming languages and almost every application uses multiple languages, the consistency of FP metric allow economic studies that are not possible using LOC metric.

Function points can be seen as a universal IT-currency converter, giving a synthetic measure of the size of software.

Q.5 What are the characteristics of good software? Explain Spiral model in detail. Name any two systems for which this model is suitable.

Ans.:- Characteristics of a good software are:-

Operational Characteristics

- Correctness: The software which we are making should meet all the specifications stated by the customer.
- Usability/Learnability: The amount of efforts or time required to learn how to use the software should be less. This makes the software user-friendly even for illiterate people.
- Integrity: Just like medicines have side-effects, in the same way a software may have a side-effect i.e. it may affect the working of another application. But a quality software should not have side effects.
- Reliability: The software product should not have any defects. Not only this, it shouldn't fail while execution.
- Efficiency: This characteristic relates to the way software uses the available resources. The software should make effective use of the storage space and execute command as per desired timing requirements.
- Security: With the increase in security threats nowadays, this factor is gaining importance. The software shouldn't have ill effects on data / hardware. Proper measures should be taken to keep data secure from external threats.
- Safety: The software should not be hazardous to the environment/life.

Transition Characteristics:

- Interoperability: Interoperability is the ability of software to exchange information with other applications and make use of information transparently.
- Reusability: If we are able to use the software code with some modifications for different purpose then we call software to be reusable.
- Portability: The ability of software to perform same functions across all environments and platforms, demonstrate its portability.

Revision Characteristics

- Maintainability: Maintenance of the software should be easy for any kind of user.
- Flexibility: Changes in the software should be easy to make.



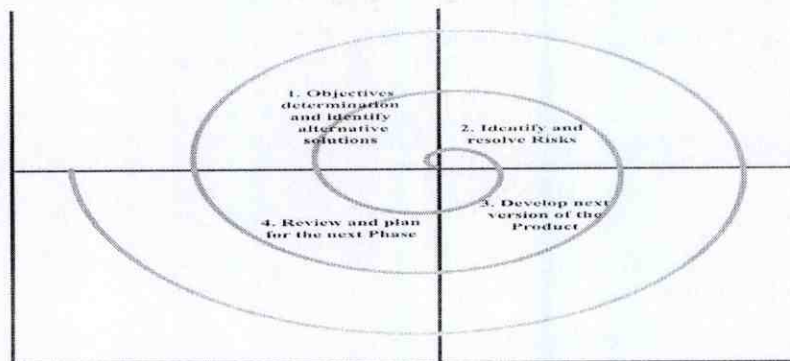
Solution of Question Paper

I Mid-Term Examination, Sept. -2022

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- c) Extensibility: It should be easy to increase the functions performed by it.
d) Scalability: It should be very easy to upgrade it for more work (or for more number of users).
e) Testability: Testing the software should be easy.
f) Modularity: Any software is said to be made of units and modules which are independent of each other.

Spiral model is one of the most important Software Development Life Cycle models, which provides support for Risk Handling. In its diagrammatic representation, it looks like a spiral with many loops. The exact number of loops of the spiral is unknown and can vary from project to project. Each loop of the spiral is called a Phase of the software development process. The exact number of phases needed to develop the product can be varied by the project manager depending upon the project risks. As the project manager dynamically determines the number of phases, so the project manager has an important role to develop a product using the spiral model.



Q.6 Describe characteristics of iterative model. Explain verification v/s validation and the different techniques of verification.

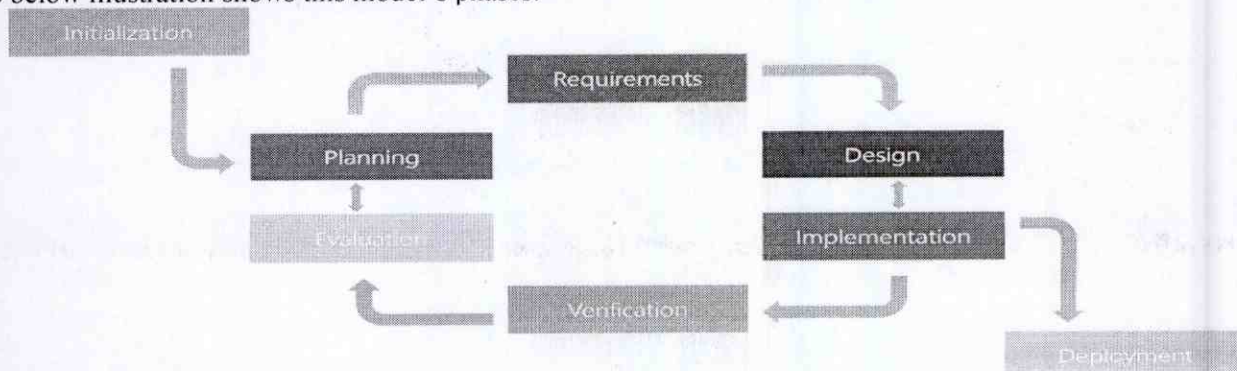
Ans.:- Iterative Model

The main objective of iterative development is to build the system incrementally, starting from basic partial system features and gradually adding more features until the entire system is completed. Compared to waterfall, iterative development allows flexibility in accommodating new requirements or changes thereof. It also provides room for improvement in succeeding iterations based on lessons learned from previous iterations.

The iterative model begins development by building only a part of the software.

Full specifications of requirements are usually not known. The small part of the system developed is reviewed to identify further requirements. Every successful iteration produces software build upon which more features are implemented. This continues until the final system is achieved.

The below illustration shows this model's phases:



Let's go through the phases for an understanding of how this model works.

Planning and Requirements:- This is the phase of collecting user requirements and expectations.

Analysis and Design:- An analysis of the requirements is done and the system is designed. Suggested solutions are discussed then one is picked. A technical representation of this solution is done.

Implementation:- The actual development happens by translating the system design into code.



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Testing:- The developed system is tested for any bugs. If found, they are fixed before deployment in the user environment.

Evaluation:- The system build is reviewed to check if it meets the specified requirements. This is also the stage where further requirements, if any, are collected.

After this point, the current build is taken back to the planning phase together with the received feedback. The whole process (iteration) begins again. This cycle continues until the system is complete.

Verification: It involves static analysis technique (review) done without executing code. It is the process of evaluation of the product development phase to find whether specified requirements meet.

Validation: It involves dynamic analysis technique (functional, non-functional), testing done by executing code. Validation is the process to evaluate the software after the completion of the development phase to determine whether software meets the customer expectations and requirements.

The different techniques of verification.

Review

- Self-review: Review done by author himself. Author knows the best about his/her code and
- Peer-review
- Superior review
- Walkthrough

Inspection:

- It is the most formal technique
- It is led by the trained moderators
- During inspection the documents are prepared and checked thoroughly by the reviewers before the meeting
- It involves peers to examine the product
- A separate preparation is carried out during which the product is examined and the defects are found
- The defects found are documented in a logging list or issue log
- A formal follow-up is carried out by the moderator applying exit criteria

PART C (Descriptive/Analytical/Problem solving/Design questions)
(Attempt any 1 Question) (1*6=6)

Q.7 Explain Agile Model in detail. Also differentiates the characteristics between Incremental Model and RAD Model.

Ans.:- Sol. Agile SDLC model is a combination of iterative and incremental process models with focus on process adaptability and customer satisfaction by rapid delivery of working software product. Agile Methods break the product into small incremental builds. These builds are provided in iterations. Each iteration typically lasts from about one to three weeks. Every iteration involves cross functional teams working simultaneously on various area

- Planning
- Requirements Analysis
- Design
- Coding
- Unit Testing and
- Acceptance Testing.



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Submitted By:- Ms. Sanju Choudhary		

Agile model believes that every project needs to be handled differently and the existing methods need to be tailored to best suit the project requirements. In Agile, the tasks are divided to time boxes (small time frames) to deliver specific features for a release.

Iterative approach is taken and working software build is delivered after each iteration. Each build is incremental in terms of features; the final build holds all the features required by the customer. Here is a graphical illustration of the **Agile Model** –

The Agile thought process had started early in the software development and started becoming popular with time due to its flexibility and adaptability.

Following are the Agile Manifesto principles –

- **Individuals and interactions** – In Agile development, self-organization and motivation are important, as are interactions like co-location and pair programming.
- **Working software** – Demo working software is considered the best means of communication with the customers to understand their requirements, instead of just depending on documentation.
- **Customer collaboration** – As the requirements cannot be gathered completely in the beginning of the project due to various factors, continuous customer interaction is very important to get proper product requirements.
- **Responding to change** – Agile Development is focused on quick responses to change and continuous development.

At the end of the iteration, a working product is displayed to the customer and important stakeholders.

RAD model distributes the analysis, design, build and test phases into a series of short, iterative development cycles.

Following are the various phases of the RAD Model –

Business Modeling

The business model for the product under development is designed in terms of flow of information and the distribution of information between various business channels. A complete business analysis is performed to find the vital information for business, how it can be obtained, how and when is the information processed and what are the factors driving successful flow of information.

Data Modeling

The information gathered in the Business Modeling phase is reviewed and analyzed to form sets of data objects vital for the business. The attributes of all data sets is identified and defined. The relation between these data objects are established and defined in detail in relevance to the business model.

Process Modeling

The data object sets defined in the Data Modeling phase are converted to establish the business information flow needed to achieve specific business objectives as per the business model. The process model for any changes or enhancements to the data object sets is defined in this phase. Process descriptions for adding, deleting, retrieving or modifying a data object are given.

Application Generation

The actual system is built and coding is done by using automation tools to convert process and data models into actual prototypes.

Testing and Turnover

The overall testing time is reduced in the RAD model as the prototypes are independently tested during every iteration. However, the data flow and the interfaces between all the components need to be thoroughly tested with complete test coverage. Since most of the programming components have already been tested, it reduces the risk of any major issues.

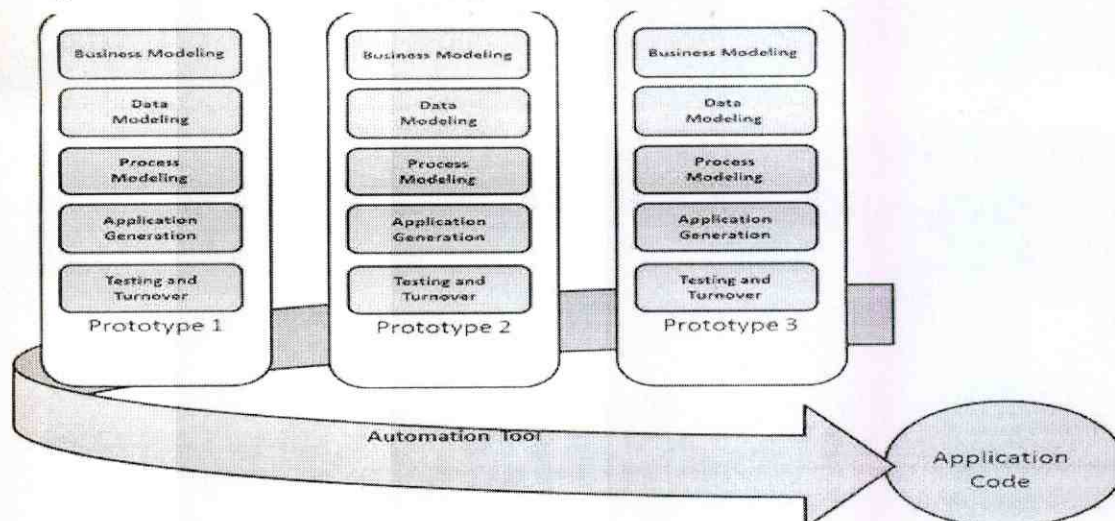


Solution of Question Paper

I Mid-Term Examination, Sept. -2022

Branch/Semester:III....	Subject:SOFTWARE ENGINEERING...	Subject Code: 3IT4-07
Duration: 1.5 hours	Date:-30/09/2022 Session:- II	Max Marks:- 20
Submitted By:- Ms. Sanju Choudhary		

The following illustration describes the RAD Model in detail.



Model Vs Traditional SDLC

The traditional SDLC follows a rigid process models with high emphasis on requirement analysis and gathering before the coding starts. It puts pressure on the customer to sign off the requirements before the project starts and the customer doesn't get the feel of the product as there is no working build available for a long time.

The customer may need some changes after he gets to see the software. However, the change process is quite rigid and it may not be feasible to incorporate major changes in the product in the traditional SDLC.

The advantages of the RAD Model are as follows –

- Changing requirements can be accommodated.
- Progress can be measured.
- Iteration time can be short with use of powerful RAD tools.
- Productivity with fewer people in a short time.

The disadvantages of the RAD Model are as follows –

- Dependency on technically strong team members for identifying business requirements.
- Only system that can be modularized can be built using RAD.

Q.8 Suppose that a project was estimated to be 800000 LOC. Calculate effort, development time, human resources and productivity for each of 3 modes of development for COCOMO Basic Model.

Ans.:- 800000 LOC = 800 KLOC

Organic:

$$\text{Effort} = 2.4(800)^{1.05} = 2681.98 \text{ person months}$$

$$\text{Time} = 2.5(2681.98)^{0.38} = 50.20 \text{ months}$$

Semi-detached:

$$\text{Effort} = 3(800)^{1.12} = 5352.81 \text{ person months}$$

$$\text{Time} = 2.5(5352.81)^{0.35} = 50.45 \text{ months}$$



**Swami Keshvanand Institute of Technology,
Management & Gramothan, Jaipur**
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Subject:	MEFA	Subject Code:	3-CE,ECE,IT,EE,ME 1-03
Time:	1.5 Hours	Maximum Marks:	20
Session (I/II/III):II			

Section A (Attempt ALL questions. Each question carries 2 marks.)

- QUESTION 1** Define the law of demand [2]
- QUESTION 2** What is demand forecasting? [2]
- QUESTION 3** Define managerial economics. [2]

Section B (Attempt ANY TWO questions. Each question carries 4 marks.)

- QUESTION 4** What is circular flow of economic activities? [4]
- QUESTION 5** Write four merits and four demerits of deductive and inductive methods. [2+2]
- QUESTION 6** Differentiate between GDP and GNP. [2+2]

Section C (Attempt ANY ONE of the following. Each question carries 6 marks.)

- QUESTION 7** What is the significance of elasticity of demand?
Explain different types of elasticity. [2+4]
- QUESTION 8** Describe six determinants of demand and supply. [3+3]



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Section A

Ans 1

The law of demand states that, if all other factors remain equal, the higher the price of a good, the less people will demand that good.

In other words, the higher the price, the lower the quantity demanded. The amount of a good that buyers purchase at a higher price is less because as the price of a good goes up, so does the opportunity cost of buying that good.

— (2)

Ans 2

Demand forecasting refers to an estimate of future demand for the product. It is an objective assessment of the future course of demand, in recent times, forecasting plays an important role in business decision – making. The survival and prosperity of a business firm depend on its ability to meet the consumer's needs efficiently and adequately. Demand forecasting has an important influence on production planning. It is essential for a firm to produce the required quantities at the right time.

— (2)

Ans 3

Economics is a study of human activity both at individual and national level. The economists of early age treated economics merely as the science of wealth. The reason for this is clear. Every one of us involved in efforts aimed at earning money and spending this money to satisfy our wants such as food, Clothing, shelter, and others. Such activities of earning and spending money are called "Economic activities".

— (1)

Managerial Economics has its roots in micro economics and it deals with the micro or individual enterprises.

It is concerned with the application of concepts such as Price Theory, Law of Demand and



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Theories of market structure and so on.

— (1)

Section B

Ans 4

The circular flow of economic activity is a model showing the basic economic relationships within a market economy. It illustrates the balance between injections and leakages in our economy. Half of the model includes injections, and half of the model includes leakages. The circular flow model shows where money goes and what it's exchanged for. The model includes households, businesses and governments. We also have the banking system that facilitates the exchange of money and, as we'll see in a minute, helps to productively turn savings into investment in order to grow the economy. In the circular flow of the economy, money is used to purchase goods and services. Goods and services flow through the economy in one direction while money flows in the opposite direction.

— (2)

The factors of production include land, labour, capital and entrepreneurship. The prices that correspond to these factors of production are rent, wages and profit. People in households buy goods and services from businesses in an attempt to satisfy their unlimited needs and wants. Households also sell their labour, land, and capital in exchange for income that they use to buy goods and services that firms produce. Businesses sell goods and services to households, earning revenue and generating profits. Businesses also pay wages, interest and profits to households in return for the use of their factors of production. Governments levy taxes on households and businesses in order to provide certain benefits to everyone.

— (2)

Ans 5

Merits of Deductive Method:



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Ans 6

Gross Domestic Product (GDP):

GDP is the total value of goods and services produced within the country during a year. This is calculated at market prices and is known as GDP at market prices.

GDP at Factor Cost:

GDP at factor cost is the sum of net value added by all producers within the country. Since the net value added gets distributed as income to the owners of factors of production, GDP is the sum of domestic factor incomes and fixed capital consumption (or depreciation).

$$\text{GDP at Factor Cost} = \text{GDP at Market Prices} - \text{Indirect Taxes} + \text{Subsidies.}$$

Gross National Product (GNP):

GNP is the total measure of the flow of goods and services at market value resulting from current production during a year in a country, including net income from abroad.

GNP includes four types of final goods and services

GNP at Market Prices.

When we multiply the total output produced in one year by their market prices prevalent during that year in a country, we get the Gross National Product at market prices.

Thus GNP at market prices means the gross value of final goods and services produced annually in a country plus net income from abroad.

$$\text{GNP at Market Prices} = \text{GDP at Market Prices} + \text{Net Income from Abroad.}$$

GNP at Factor Cost.

GNP at factor cost is the sum of the money value of the income produced by and accruing to



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1. Real:
2. Simple:
3. Powerful:
4. Exact:
5. Indispensable:
6. Universal:

— (1)

Demerits of Deductive Method:

1. Unrealistic Assumption:
2. Not Universally Applicable:
3. Incorrect Verification:
4. Abstract Method:
5. Static Method:
6. Intellectually:

— (1)

Merits of Inductive Method:

1. Realistic:
2. Future Enquiries:
3. Statistical Method:
4. Dynamic:
5. Historic-Relative:

— (1)

Demerits of Inductive Method:

1. Misinterpretation of Data:
2. Uncertain Conclusions:
3. Lacks Concreteness:
4. Costly Method:
5. Difficult to Prove Hypothesis:
6. Controlled Experimentation not Possible in Economics:

— (1)



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the various factors of production in one year in a country. It includes all items mentioned above under income method to GNP less indirect taxes.

GNP at Factor Cost = GNP at Market Prices – Indirect Taxes + Subsidies.

Section C

Ans 7

Elasticity of Demand:

Elasticity of demand explains the relationship between a change in price and consequent change in amount demanded. "Marshall" introduced the concept of elasticity of demand. Elasticity of demand shows the extent of change in quantity demanded to a change in price.

In the words of "Marshall", "The elasticity of demand in a market is great or small according as the amount demanded increases much or little for a given fall in the price and diminishes much or little for a given rise in Price".

Elastic demand:

A small change in price may lead to a great change in quantity demanded. In this case, demand is elastic.

Inelastic demand:

If a big change in price is followed by a small change in demanded then the demand is "inelastic".

Types of Elasticity of Demand:

There are four types of elasticity of demand:

1. Price elasticity of demand:

$$\text{Price elasticity} = \frac{\text{Proportionate change in the quantity demand of commodity "X"}}{\text{Proportionate change in the price of commodity "X"}}$$

$$\text{Edp} = \frac{(Q_2 - Q_1) / Q_1}{(P_2 - P_1) / P_1}$$



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2. Income elasticity of demand:

$$\text{Income Elasticity} = \frac{\text{Proportionate change in the quantity demand of commodity}}{\text{Proportionate change in the income of the people}}$$

$$EdI = \frac{(Q_2 - Q_1)/Q_1}{(I_2 - I_1)/I_1}$$

— (1)

3. Cross elasticity of demand:

$$\text{Cross Elasticity} = \frac{\text{Proportionate change in the quantity demand of commodity "X"}}{\text{Proportionate change in the price of complementary commodity "Y"}}$$

$$Edc = \frac{(Q_2 - Q_1)/Q_1}{(P_2 - P_1)/P_1}$$

— (1)

4. Advertising elasticity of demand:

$$\text{Advertising Elasticity} = \frac{\text{Proportionate change in the quantity demand of commodity}}{\text{Proportionate change in advertisement costs.}}$$

$$EdA = \frac{(Q_2 - Q_1)/Q_1}{(A_2 - A_1)/A_1}$$

— (1)

Ans 8

Determinants of Demand:

There are so many factors on which the demand for a commodity depends. These factors are Economic, Social as well as Political factors. The affect of all these factors on the amount of demanded for the commodity is called Demand Function. The following are some of the factors that cause a change in demand other than price factor.



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1. Price of the Commodity:

The most important factor affecting on demand is the price of the commodity. The amount of the commodity demanded at a particular price is more popularly called price demand.

2. Prices of Related Goods

Change in the Prices of Substitutes: In case of substitutes like tea and coffee an increase in price of one commodity leads to an increase in the demand for other commodity and vice versa. The rise in price of coffee shall raise the demand for tea. — (1)

3. Income of the Consumer:

The third most important factor influencing demand is consumer income. In fact we can establish a relationship between the consumer income and demand at different levels of income, price and other things remaining same

4. Tastes and Fashions of Consumers:

The fourth most important factor influencing demand is consumers' tastes and fashions. The demand also depends on consumer's taste. Tastes include fashion, habit, customs etc. A customer taste is also affected by advertisement — (1)

5. Affect of Wealth:

The amount demanded of the commodity is also affected by the amount of wealth as well as its distribution. When the wealth of the people is more, demand for the normal commodities is also more.

6. Change in Population:

Increase in population increases demand for necessities of life. The compositions of population also affect demand. Composition of population means the proportion of young and old and children as well as the ratio of men and women — (1)

Determinants Of Supply

1. The cost of factors of production:

Cost depends on the price of factors. Increase in factor cost increases the cost of production, and reduces supply.

2. The state of technology:

Use of advanced technology increases productivity of the organization and increases its



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supply.

3. External factors:

External factors like weather influence the supply. If there is a flood, this reduces supply of various agricultural products.

4. Tax and subsidy:

Increase in government subsidies results in more production and higher supply.

5. Transport:

Better transport facilities will increase the supply.

6. Price:

If the prices are high, the sellers are willing to supply more goods to increase their profit.